Investment Tax Credits and the Response of Firms*

Adrian Lerche†

Universitat Pompeu Fabra

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Abstract

This paper estimates the causal impact of investment tax credits on input choices of firms and quantifies the indirect contribution of regional spillovers. For identification, I exploit a shift in the tax credit rate of small compared to large manufacturing firms in economically lagging regions of Germany in 1999. I find that tax credits lead to an increase in both investment and the number of employees of firms. While there is little evidence that the average firm adjusts its skill mix or occupation structure, firms in industries with a larger share of information and communication technology (ICT) are more likely to shift towards high-skilled labor and occupations. The employment effect materializes almost exclusively through the hiring of additional employees rather than increased separations and the unemployed constitute a considerable share among the new hirings. Regional spillovers lead to an additional positive employment effect.

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Providing tax relief is an integral part of fiscal policies for improving economic outcomes. Governments have long used temporary tax reductions for firms and frequently discuss long-term corporate tax code changes. Starting with Hall and Jorgenson (1967) a large literature emerged on the effect of tax incentives on firm investment. Empirical research surveyed in Hassett and Hubbard (2002) and subsequent studies suggest that

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†Universitat Pompeu Fabra, Ramon Trias Fargas 25-27, 08005 Barcelona, Spain. E-mail: adrian.lerche@upf.edu

there is indeed an investment response. However, tax incentives are implemented to generate broad welfare benefits throughout an economy and particularly for the labor force. But so far there is little evidence on the adjustment behavior of firms with respect to their labor input choices.

In this paper, I investigate the effect of investment tax credits on both firm investment and employment. To better understand the adjustment for different labor types, I analyze the effect of tax credits on the hiring of unemployed individuals and the shift of firm composition towards high-skilled labor due to capital-skill complementarity (Krusell et al., 2000). Furthermore, I consider agglomeration economies for creating employment through indirect channels.

I study a large-scale government program in Germany which started in 1991 directly after German reunification. Due to considerable differences in the economic development during separation, the reunited country consisted of disparate regions with East Germany having much lower production efficiency and higher unemployment. To achieve convergence the German government provided investment incentives to firms in these lagging regions by giving tax credits as a share of investment costs. The financial relief was substantial. Tax credits reduced investment costs by up to 27.5% and amounted to additional government expenses of €1-2 billion per year.

For identification I exploit variation in the tax credit rate with respect to firm size and over time. During the period of analysis manufacturing firms with at most 250 employees received higher tax credits. In 1999 an update of the rates amplified this differential treatment in favor of firms below the cutoff. I implement a difference-in-differences design comparing the behavior of firms below and above the cutoff, before and after the policy change. I exclude observations close to the cutoff to avoid interference from bunching behavior. In a second step, I use variation on the labor market level to estimate spillover effects. A simple theoretical framework shows how the policy change creates both firm level and regional variation and how this allows to separately estimate direct firm adjustments and indirect spillover effects.

The analysis relies on two distinct administrative datasets that both collect panel information on German firms. A dataset from the Federal Statistical Office of Germany provides information on the universe of manufacturing firms with more than 20 employees since 1995 and contains output and detailed firm investment information. For a better

understanding of changes in employment, I use a 50% random sample of all German firms for information by skill and occupation, and employee inflows and outflows from the Institute for Employment Research (IAB).

My first empirical finding is that tax credits have a substantial effect on investment. The increase in the relative tax credit rate leads to 0.234 log points higher overall investment and 0.251 log points higher equipment investment. This investment response corresponds to an intensive margin semi-elasticity with respect to the tax credit rate of 3.3 and is similar to the large response found in Zwick and Mahon (2017). There is however no discernible effect on the probability of investing. Looking at the dynamic effect, the intensive margin investment response picks up directly after the policy change. The effect gets stronger in the subsequent year and stabilizes at that higher level.

I then consider labor inputs and I find a positive effect of tax credits on overall employment within firms. With the preferred estimate the tax credit rate change leads to 0.087 log points higher employment. This effect is similar when considering full-time employees, regular employees or the subset of both. Most of the effect materializes already in the year after the policy change and there is a slight further increase in subsequent years. These adjustments in inputs then translate to more output, measured in terms of revenue. Estimates however tend to be more volatile and less significant. The response to the tax credit rate change for domestic revenue is 0.083 log points.

To verify that these results are not driven by the research design or its implementation, I consider various robustness checks. First, by plotting the raw data year by year and reconsidering the dynamic estimation, I find no differential pre-treatment behavior between treatment and control group. Furthermore, adjustments in inputs are apparent directly in the year after the policy change. Second, the results are robust to different sample selection. Average effects are stable to selecting narrower or wider firm size intervals and to excluding different bandwidths of firms around the cutoff. Third, using only treated firms and assuming various policy irrelevant firm size cutoffs produces insignificant results. All main regressions and robustness checks also include industry-year and labor market-year fixed effects to control for time-varying industry shocks and differential regional development.

Taking these results together, tax credits are an effective method to induce higher investment and employment in firms. In the subsequent analysis, I focus on labor in-

puts and look at different dimensions of adjustment. First, I find that the increase in employment happens almost exclusively through hiring of additional employees. There is also an increase in separations for the same firms, although estimates are small and statistically insignificant. This points to adjustments in the overall employment structure and counters the concern that the employment effect materializes through reallocation of employees from control to treatment group. Among newly hired employees, a share of 49% is previously unemployed which is similar to the share among all hirings.

Considering capital-skill complementarity, the employment effect might mask important heterogeneity by skill. I compare college educated, high-skill occupations, manual occupations and withing manual occupations those that are skilled, with their counterparts. All point estimates are close to zero and statistically insignificant. Standard errors reject all but modest skill composition changes. Thus, the added capital in firms does not shift employment opportunities towards high-skilled or service jobs. The literature considers information and communication technology (ICT) as particularly skill biasing. When looking at heterogeneous effects on industry level by share of ICT in investment and capital, I find that industries with larger shares are more likely to shift towards college educated employees and high-skilled occupations.

These results so far all speak to the direct effect of tax credits. By exploiting variation on a labor market level, I shift the focus to indirect spillover effects. I find that firms in labor markets with a larger share of firms below the cutoff experience a larger employment effect. When adjusting the shares by input-output linkages between industries, there is a similar response. These findings suggest an additional indirect effect through regional spillovers where input-output linkages play an important role.

I follow a long literature that quantifies the effect of capital cost changes on firm investment in general (Hall and Jorgenson, 1967; Abel, 1980; Summers, 1981; Abel and Blanchard, 1986) and by using changes in tax policy (Cummins et al., 1994; Goolsbee, 1998; House and Shapiro, 2008; Edgerton, 2010; Yagan, 2015; Maffini et al., 2016; Zwick and Mahon, 2017). Early studies heavily rely on a structural approach and find surprisingly small responses in investment.[1] Strong assumptions about the capital formation process and limited variation in capital costs may be the reason for these findings since the same estimation approach using tax policy changes leads to much larger effects. The

[1]The elasticities are considered *too* small, since they imply extremely high capital adjustment costs.

recent literature shifts towards the analysis of specific tax incentive programs that introduce cross-sectional variation and finds even larger responses in capital. Evidence suggests that heterogeneous effects by firm size and profitability can explain some of these differences at least in the case of special depreciation allowances. Yagan (2015) does not find any effect, however, he analyzes a dividend tax cut in the U.S. that may be quite distinct from tax policies targeting investment directly.

I contribute to this literature by studying firm investment in a novel setting, using the tax credit program in Germany, which broadens the understanding of different types of tax policies and relates results to a new institutional setting. More importantly, I go beyond the analysis of investment and focus on labor input choices in the context of tax policies. My findings help to assess the benefits of tax policies, since increases in employment are an important reason for implementing them in the first place. I furthermore explore differential adjustments by labor type which is important for the understanding of heterogeneous welfare effects. The analysis is facilitated by the particular setup of the tax credit program. Given the simple firm size cutoff, I can use datasets that have more detailed employment information than is available from tax reports. The classification of treatment and control group by firm size also allows to directly control for industry and region specific shocks.

With this broader focus, my paper also connects to the large literature on evaluating place-based policies both on a regional level (Becker et al., 2010, 2013; Busso et al., 2013; Kline and Moretti, 2014a; Criscuolo et al., 2016) and firm level (Bronzini and de Blasio, 2006; Cerqua and Pellegrini, 2014) that already puts a stronger focus on employment. In these studies, the effect on employment is mixed which may be because of the differences in the mix of regional and firm specific incentives of each place-based policy. I add to this literature by looking at one particular incentive, investment tax incentives, and study adjustments in input and output independent of other influences. By considering heterogeneous effects by labor types, I also speak to the often voiced concern in this literature (Glaeser and Gottlieb, 2008; Kline and Moretti, 2014b; Neumark and Simpson, 2015) that such policies can lead to unwanted redistribution in welfare. The analysis by skill type further relies on the insides from the vast literature on the shift of production technology towards automation (e.g. Acemoglu and Autor, 2011) and capital in general (Krusell et al., 2000). Similar to Akerman et al. (2015), my results address potential

adverse effects of government programs in the context of ICT.

In addition, my results highlight the importance of agglomeration economies as indirect channel in tax policies. I introduce a simple theoretical framework that separates direct and indirect effects in the context of the tax credit program. This approach is similar to Greenstone et al. (2010) and Gathmann et al. (2017) which examine exogenous variation in firm openings and closings respectively, and find important regional spillover effects.[2] Agglomeration economies are one mechanism that is usually left out in firm level studies, but is part of effects found in macro-level time series studies (Mertens and Ravn, 2013).[3]

The structure of the paper is as follows. Section 1 explains the policy intervention in more detail and focuses on the specific regulations that are important for the empirical analysis. Section 2 introduces a theoretical framework that provides intuition for expected firm behavior with heterogeneous labor types and agglomeration economies. Section 3 explains the estimation strategy. Section 4 provides details on the data including descriptive statistics and sample selection. Section 5 presents the main results and Section 6 concludes.

1 Policy Intervention

After the second world war, Germany split into two countries, West Germany and East Germany.[4] While West Germany experienced continued growth with a market-based economy, East Germany stagnated due to large war reparations and inefficiencies in their communist economical system. In 1990 the communist system collapsed and Germany reunited. The diverging prior development however created a country with economically disparate regions. For 1991 to 1994, East Germany had on average 46% lower GDP per capita, 47% lower capital per worker, 30% lower earnings per worker and an unemployment rate of 13.4% compared to 7.1% in West Germany (see Figure 1). To achieve convergence the government provided considerable financial support to regions in East

[2]The overall literature on agglomeration economies is much larger. For an overview see Combes and Gobillon (2015).

[3]Various other time series studies like Blanchard and Perotti (2002) and Romer and Romer (2010) do not consider employment but look at output as main variable of interest.

[4]During the time of separation the official designation for West Germany was *Federal Republic of Germany* and for East Germany *German Democratic Republic*. I use the common names since they still have meaning for referring to the parts of Germany after reunification.

Germany. Besides cash transfers to private households and large infrastructure investments, efforts were focused on increasing the capital stock of firms. The most salient policy in this respect was an investment tax credit program (*Investitionszulagengesetz*) which is at the center of this paper.

The program started almost immediately after reunification in 1991 and lasted until 2013. It provided tax credits for equipment investment to establishments located in East Germany and West Berlin.[5] Since 1999, it also covered structures investment. At the beginning of the program, firms of all industries were included, but over time industries lost eligibility and by 1997 coverage applied almost exclusively to manufacturing firms.[6] Tax credits typically reduced investment cost by around 10% but in some instances the reduction was as high as 27.5%.[7] Tax credits were fully paid even if they exceeded tax liabilities of a firm, and they did not depend on the life span of the investment good.[8]

Importantly, all details on firm eligibility and tax credit rate were precisely defined by law without room for discretion on a case by case basis. This led to an entitlement to tax credits for eligible firms and thus certainty for planning investment projects.[9] The tax credits therefore can be considered a pure capital cost reduction for firms. The administrative cost for receiving tax credits was small. Firms filled out a tax credit claim form describing the investment good and the value of investment. Tax officers checked the correctness of the claim after the end of the business year and a positive assessment would trigger the transfer of tax credits. Further eligibility criteria attempted to reduce adverse incentives. Assets had to stay within the firm for at least 3 years, likely to avoid that East German firms became pass-through companies of buying and reselling fixed

[5]The differentiation between firm and establishment in this context is important. Firms with establishments in West and East were eligible for grants as long as the investment pertained to establishments in the East. For the empirical analysis, I consider establishment and firm to be the same unit and refer to them as "firm". I will show evidence that the discrepancy of this simplification for multi-establishment firms is negligible.

[6]Retail businesses continued having limited eligibility until 2001. Manufacturing service businesses such as construction design or research gained access to tax credits in 1999. Accommodation businesses received eligibility starting in 2007.

[7]This highest rate applied to equipment investments of manufacturing firms with 250 or fewer employees in regions close to the Czech and Polish border from 2002 to 2009.

[8]This is in contrast to special depreciation allowances where the decrease in cost of capital depends on the profit situation of a firm and on the years of depreciation due to the present value of future tax deductions. House and Shapiro (2008) and Zwick and Mahon (2017) provide detailed explanations.

[9]This is in contrast to various place-based policies that distribute grants to investment projects via a competitive application process with the final outcome based for example on the perceived success of the project or a commitment to hire additional employees.

assets.[10] Planes, passenger cars and low value assets such as office equipment or basic tools never received eligibility, because the rightful use within an eligible firm in East Germany (or West Berlin) would entail large monitoring costs.

Figure 2 summarizes government expenses for the tax credit program by year. From 1992 to 1995 expenses were summing up to around €2 billion per year. After 1993, expenses declined steadily, which can be explained with the narrowing of eligible industries. They reached a low of €645 million in 1998 and stabilized thereafter at around €1 billion. Starting in 2000, expenses from tax credits for structures investments added to the total and they generally made up around 15% each year.

In the empirical analysis, I focus on manufacturing firms as the main benefiters of the tax credit program and consider the time period between 1995 and 2004 to compare their behavior before and after an important program update to tax credit rates for equipment investments in 1999. Starting in July 1994 firms in the manufacturing industry with 250 employees or less received a tax credit of 10% on equipment investments. Firms with more employees instead received 5%. For the considered time period the program defined firm size according to the number of employees at the beginning of a business year without differentiating full-time and part-time employment. Vocational trainees were excluded from the definition since their employment is considered educational and is governed by special contracts. The according tax credit rate applied to all investments of the same business year. In 1999, tax credits were raised for a broad range of equipment investments. From then on firms below the employment cutoff received tax credits of 20% whereas those above received a rate of 10%.[11] This adjustment led to higher tax credits for all manufacturing firms. However, firms below the cutoff were favored with an increase of 10 percentage points compared to 5 percentage points which translates to a larger relative decrease of capital costs of equipment for those below the cutoff.[12]

The change applied to so called modernization investments, which included among others any investment that could potentially increase production, change the production

[10] After 1999 the minimum time period was 5 years.

[11] The announcement of the policy update was published in August 1997 (Bgbl, 1997 Part I/59). Because of EU law, important parts of the update got approval only by the end of 1998 (European Commission Press Release IP/98/1075). At the same time an investment limit for receiving the higher tax credit rate of €2.56 million per year for firms below the employment cutoff was eliminated.

[12] The introduction of tax credit rates for structures investment coincided. However, the rate was independent of firm size and thus reduced capital costs of structures equally. Appendix A provides details on similar issues and how the unique characteristics of the policy change in 1999 can identify tax credit rate changes.

process or produce different products. Any investment in new machinery that did not immediately replace a similar asset fell within this category. Even (high value) office equipment could be part of this category as long as they were bought in connection to a specific modernization investment.[13]

Figure 3 summarizes the general tax credit changes for equipment investments of manufacturing firms in East Germany between 1995 and 2004.[14] Apart from the adjustment in 1999, there was a further increase for modernization investments in 2000 that strengthened the relative advantage of firms below the cutoff to those above. Tax credits for non-modernization equipment investment stayed unchanged during the policy update in 1999 but were reduced in 2002. However, this reduction did not change the differential treatment and maintained a higher rate of 5 percentage points for firms below the cutoff before and after the change.

Since the definition for firm size changed markedly in 2005, I exclude those years in my analysis. From then on the cutoff value followed the definition of small and medium firms by the European Union that takes ownership structure into account and defines the cutoff along the number of employees, revenue and total assets.[15]

2 Theoretical Framework

To understand firm behavior after capital cost changes, it is helpful to outline a simple firm production model. The literature already developed a detailed understanding of firm investment behavior.[16] In these models capital adjustment costs and corporate taxation are an integral part. I abstract from both these issues and focus on a static model with capital costs consisting of a universal capital rental rate and a firm-specific tax credit rate.[17] Nevertheless, I keep labor as input to the production process and

[13]Further explanations about the categorization can be found in the remarks from a meeting of the German Ministry of Finance with regional tax authorities on 25-27 April 2001 and subsequent similar meetings.

[14]Berlin had generally lower rates throughout the time period. Regions close to the Polish and Czech border received slightly higher rates between 2001 and 2009. For long investment projects, additional rules applied around changes of the tax credit rate.

[15]The definition can be found in the "Commission recommendation of 6 May 2003 concerning the definition of micro, small and medium-sized enterprises" (ABl.EU, #L 124 pp36).

[16]Important examples are Hall and Jorgenson (1967), Hayashi (1982) and Abel and Eberly (1994). Hassett and Hubbard (2002) and Bond and Reenen (2007) summarize basic model assumptions and survey further approaches.

[17]The qualitative equilibrium results hold true nonetheless. An important difference is that in my model firms choose capital stock and not the investment rate as is the case with adjustment costs.

consider heterogeneous labor types. I also add a regional productivity shifter similar to Greenstone et al. (2010) and Gathmann et al. (2017). By doing so, the model provides predictions for firm adjustment of labor and captures the influence of spillovers in regional firm production.[18]

The model assumes many firms i within many regions r. Each firm produces one differentiated good according to the nested CES production function

$$F(K_i, U_i, S_i) = Y_i = A_i A_r \left[(a_K K_i^{\rho} + a_S S_i^{\rho})^{\frac{\mu}{\rho}} + a_U U_i^{\mu} \right]^{\frac{1}{\mu}}, \tag{1}$$

where output Y_i is produced using capital K_i, low-skilled labor U_i and high-skilled labor S_i as inputs. The nesting of the three input factors follows Krusell et al. (2000) to allow for differential adjustment of the two labor inputs to a change in capital costs. In particular, the elasticity of substitution between low-skilled labor and capital (or high-skilled labor) is $\frac{1}{1-\mu}$ and the elasticity of substitution between high-skilled labor and capital is $\frac{1}{1-\rho}$. Production also depends on a firm-specific production parameter A_i and a region-specific productivity shifter A_r. I consider the behavior of A_r in detail once I turn to the effect of spillovers.

Each firm chooses inputs according to the rental rate of capital r, wage w_U for low-skilled and wage w_S for high-skilled. I assume fully elastic labor supply and capital supply which leads to equalization of input prices throughout the economy. Firms take the input prices as given. The cost of capital still differs between firms since there is a firm-specific reduction through tax credits with rate τ_i. Firms set the product price p_i facing monopolistic competition with a downward sloping inverse demand curve

$$p_i = B Y_i^{-\frac{1}{\eta^D}}, \tag{2}$$

where the price depends on the elasticity of demand $\eta^D > 1$ and a demand shifter B.[19]

The profit maximization problem for firms is well defined and the first-order conditions

[18] Details on solving the model are provided in Appendix B.

[19] Monopolistic competition leads to decreasing returns to scale. In this case, firms output is always well-defined. Other approaches introduce a fixed input factor or restrict full elasticity of labor supply.

fully explain the production decisions of a firm. The first-order conditions are

$$(1-\tau_i)r = \left(1-\frac{1}{\eta^D}\right) Ba_K Y_i^{1-\mu-\frac{1}{\eta^D}} X_i^{\frac{\mu-\rho}{\rho}} K_i^{\rho-1}(A_iA_r)^{\mu} \tag{3}$$

$$w_S = \left(1-\frac{1}{\eta^D}\right) Ba_S Y_i^{1-\mu-\frac{1}{\eta^D}} X_i^{\frac{\mu-\rho}{\rho}} S_i^{\rho-1}(A_iA_r)^{\mu} \tag{4}$$

$$w_U = \left(1-\frac{1}{\eta^D}\right) Ba_U Y_i^{1-\mu-\frac{1}{\eta^D}} U_i^{\mu-1}(A_iA_r)^{\mu} \tag{5}$$

where $X_i = a_K K_i^{\rho} + a_S S_i^{\rho}$.

To show the impact of a change in cost of capital through adjustments in the tax credit rate on optimal capital and labor, I totally differentiate the production function (1) and all first order conditions (3), (4) and (5). By reformulating the results, the price elasticity of capital and the cross-price elasticity of high-skilled and low-skilled labor respectively are

$$e_{K_i} = \left[\eta^D s_K + \frac{1}{1-\mu}s_U + \frac{1}{1-\rho}s_S + \left(\frac{1}{1-\rho}-\frac{1}{1-\mu}\right)s_U\frac{a_S S_i^{\rho}}{X_i}\right]Z \quad + \quad (\eta^D-1)e_{A_r} \tag{6}$$

$$e_{S_i} = \left[\left(\eta^D-\frac{1}{1-\rho}\right)s_K + \left(\frac{1}{1-\mu}-\frac{1}{1-\rho}\right)s_U\frac{a_K K_i^{\rho}}{X_i}\right]Z \quad + \quad (\eta^D-1)e_{A_r} \tag{7}$$

$$e_{U_i} = \underbrace{\left(\eta^D-\frac{1}{1-\mu}\right)s_K Z}_{\text{direct effect}} \quad + \quad \underbrace{(\eta^D-1)e_{A_r}}_{\text{indirect effect}}, \tag{8}$$

where each elasticity $e_{\#} = \frac{d\#}{d\tau_i}\frac{1-\tau_i}{\#}$ is with respect to the tax credit rate,[20] $s_K = \frac{(1-\tau_i)r}{p_i}\frac{K_i}{Y_i}$ is the share of capital cost (after tax credit) in revenue, $s_U = \frac{w_U}{p_i}\frac{U_i}{Y_i}$ is the share of low-skilled labor cost in revenue, $s_S = \frac{w_S}{p_i}\frac{S_i}{Y_i}$ is the share of high-skilled labor cost in revenue and $Z = \frac{\eta^D}{\eta^D-1}$ is an additional scaling term.

Each elasticity consists of a firm-specific direct effect and an indirect regional effect. Turning first to the direct effect of tax credits, the elasticities resemble the text book case (e.g. Hamermesh, 1993) quite well. The elasticity of capital and labor largely depend on two effects, a scale effect from changes in input prices and a substitution effect between capital and the two types of labor. In the case of capital, these effects work in the same direction. An increase in tax credits leads to more demand in capital because of an expansion of production and a shift from labor towards capital. On the other hand, for each labor type the overall effect is ambiguous since there is higher demand

[20] It is straightforward to show that this elasticity is equivalent to $-\frac{d\#}{dc_K}\frac{c_K}{\#}$, the capital cost elasticity.

from expansion in production but lower demand from the shift towards capital. The total depends importantly on the elasticity of product demand and the elasticity of substitution of the according labor type. Because of monopolistic competition and the nesting of the CES production function, additional terms show up. First, less than fully elastic product demand amplifies adjustments of all inputs through the term Z. Second, an additional term leads to different weighting of each substitution effect for the elasticity of capital. Third, there is also a change in weighting for the elasticity of high-skilled labor and there is an additional influence from the elasticity of substitution of low-skilled labor.

To better judge the effect on composition of labor input, it is useful to look at the elasticity of the ratio between high-skilled and low-skilled with respect to the tax credit rate. It is

$$\frac{d\frac{S_i}{U_i}}{d\tau_i}\frac{1-\tau_i}{\frac{S_i}{U_i}} = \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)\frac{a_K K_i^{\rho}}{X_i}. \tag{9}$$

The values depends on the size of each elasticity of substitution. If the elasticity of low-skilled labor is higher than the one of high-skilled, there is capital-skill complementarity with a shift towards high-skilled labor.

Turning to the indirect effect, a change in the regional productivity shifter influences each elasticity of input equally. The indirect effect even applies to firms without change in their capital costs as long as they are located in the same region. The elasticities of firm j with respect to the tax credit of firm i are

$$\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} = \frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j} = \frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} = (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \tag{10}$$

I follow the literature and assume that the productivity shifter depends on the overall economic activity within a region

$$A_r = \left(\sum_{j \in S_r} Y_j\right)^{\lambda}, \tag{11}$$

where the set S_r contains all firms in region r and λ is the elasticity of agglomeration.[21] The productivity shifter encompasses all sources of agglomeration economies discussed in the literature such as cost savings form proximity in production networks, localized

[21] I use output as measure for economic activity since it is unambiguously positive for tax credit rate increases.

knowledge spillovers and thick labor markets (e.g. Moretti, 2011).

Using this definition of regional productivity and the elasticity of output, the according elasticity of regional productivity with respect to a change in tax credit rate of one firm is

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \lambda\frac{1-\tau_i}{\sum Y_j}\sum_j \frac{dY_j}{d\tau_i} = \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{Y_i}{\sum Y_j}s_{K_i}Z > 0, \tag{12}$$

where s_{K_i} is the share of capital cost in revenue for firm i and I assume that $\lambda\eta^D < 1$.

The elasticity importantly depends on the interaction of the elasticity of agglomeration and the elasticity of product demand. The larger both, the more pronounced the impact on regional productivity. Furthermore, firms with a higher share of output in total regional output and larger shares of capital cost in their revenue impact regional productivity more. An increase in tax credit rate always leads to a positive spillover effect. The additional assumption on the parameters excludes all boundary cases in which spillovers are so large that total output within a region becomes infinite.

If several firms receive a change in tax credits, then the effect on regional productivity gets larger since more firms increase their output through the direct channel. Intuitively, the elasticity with respect to the tax credit rate of multiple firms comes about by sequentially calculating the equilibrium adjustments. Considering infinitesimally small tax rate changes, this reduces to summing up all elasticities of regional productivity for firms with changing capital costs. As result, the elasticity is

$$\frac{d\ln A_r}{d\ln \boldsymbol{\tau}} = \sum_{j|d\tau_i\neq 0}\frac{dA_r}{d\tau_j}\frac{1-\tau_j}{A_r} = \frac{\lambda\eta^D}{1-\lambda\eta^D}Z\frac{\sum s_{K_i}Y_{i|d\tau_i\neq 0}}{\sum Y_j}. \tag{13}$$

The elasticity has a very similar form. However, having multiple firm adjustments leads to a summation of the output of all firms with tax credit rate change weighted by their share of capital costs. Regions where more firms experience a rate change end up with larger effects from spillovers.

3 Estimation Strategy

The estimation strategy is guided by the theoretical framework and uses the described update of tax credit rates in 1999. I implement a difference-in-differences estimation approach and compare the adjustment of firms below and above the firm size cutoff

before and after the change. I start with the estimation of average effects using the regression model

$$Y_{ibt} = \beta \text{Treated}_b \times \text{Post}_t + X_{it}^{'}\gamma + \psi_i + \psi_{nt} + \psi_{rt} + \epsilon_{ibt}, \tag{14}$$

where Y_{ibt} is the outcome variable for each firm i with treatment b in year t. The variable Treated_b classifies firms into treatment and control group. I consider firms with at most 250 employees in 1998 as treated and those above as untreated to reflect the relative advantage for firms below the cutoff. The classification is fixed over time. I interact this variable with the dummy variable Post_t, which categorizes years 1999 to 2004 as treatment period to reflect the change in tax credit rate. ψ_i, ψ_{nt} and ψ_{lt} are firm, industry-year and labor market-year fixed effects, respectively. Firm fixed effects control for level differences in firm characteristics that stay constant over time such as average firm size, industry and location. Industry-year and labor market-year fixed effects can control for the possibility that industry-specific shocks and changes between regions coincide with the update of the tax credit program. I include further control variables $X_{it}^{'}$ depending on the specification with log firm wage being used for all main results.[22]

The coefficient of interest is β. Without confounding factors, it provides a causal estimate for the effect of having lower capital costs on each outcome variable, using the differential tax credit rate adjustment between firms below and above the cutoff after the policy change. The set of fixed effects already controls for various confounding factors. However, it cannot take care of distinct firm incentives that are introduced by the firm size cutoff. Firms can adjust their size over time and thus are able to cross the cutoff. Such movements imply that these firms receive a different tax credit rate than assigned by the treatment status in the regressions. If this change in firm size is unrelated to the tax credit program, for example because of a general decline of a specific firm, this would not influence the causality claim, although it influences the interpretation of the coefficients. To minimize this issue, I exclude firms that are close to the cutoff in 1998 and thus those that are more likely to cross the cutoff in either way. On the other hand, if firms intentionally move just below the cutoff or delay moving above, this would result in bunching at the cutoff. To avoid such behavior in the estimation, I further exclude observations in any year for which firm size is close to the cutoff and thus potentially

[22] I also look at pre-treatment wage growth trends for employment outcomes and find similar results.

affected by bunching.

I look at several outcomes. First, I am interested in the effect on capital input factors. As is common in the literature, I use investment as a directly measured variable in the dataset to proxy capital adjustment. I consider the intensive margin, extensive margin and a combined measure. I also specifically consider equipment investment. In a second step, I focus on labor inputs and analyze the effect on the number of total employees and for various subcategories separately. For a better understanding of skill composition, I consider employment ratios by education and occupation.

The error term ϵ_{ibt} includes all other omitted factors. I cluster standard errors on a regional level allowing heteroscedasticity and arbitrary correlation between firms within the same region over time.[23]

Since the average effect can mask interesting adjustment patterns, I estimate the dynamic regression model

$$Y_{it} = \sum_{p=1995}^{1997} \delta_p D_{bp} + \sum_{p=1999}^{2004} \delta_p D_{bp} + X_{it}^{'}\gamma + \psi_i + \psi_{nt} + \psi_{lt} + \epsilon_{ibt} \tag{15}$$

where in comparison to Specification (14) the interaction term is omitted and instead a set of yearly dummies D_{bp} is introduced. Each dummy variable assigns a value of one to firms belonging to the treatment group for the according year p and zero otherwise. I include dummies for the years 1999 to 2004 to estimate the dynamic treatment effect. As identifying assumption in a difference-in-differences estimation, treatment and control group must move in parallel absent the policy change. This might not be the case because of anticipation effects from the announcement of the policy or long-term influences from similar policy changes before 1995. I include dummies for the years 1995 to 1997 and examine the pre-treatment behavior. Having statistically insignificant estimates close to zero before the treatment begin provides some indication that the identifying assumption indeed holds true.

Both regression approaches so far only consider the direct effect of the policy change. For the estimation of spillover effects, I use the regression model

$$Y_{iblt} = \beta \text{Treated}_b \times \text{Post}_t + \alpha \text{ShareBelow250}_{l,98} \times \text{Post}_t + X_{it}^{'}\gamma + \psi_i + \psi_t + \epsilon_{iblt}, \tag{16}$$

[23] I consider German *Landkreise* as regions in my analysis.

where $\text{ShareBelow250}_{l,98}$ is the share of firms with at most 250 employees in labor market l for year 1998, weighted by number of employees. The variable is interacted with the treatment period dummy Post_t to set up a difference-in-differences estimation with continuous treatment status. There are now two coefficients of interest, β, the direct effect of receiving a higher tax credit rate as in Specification (14) and α. The latter coefficient provides an estimate of the difference in the outcome variable by the share of firms below the cutoff that is due to the change in tax credit rate. This setup mimics the results of the theoretical framework and therefore α corresponds to an estimate of agglomeration economies. Since I use variation on a labor market level for this estimation, I cannot control for the same set of fixed effects. Instead, I include firm and year fixed effects and the log of firm wage as baseline controls.

The analysis of dynamic effects is again helpful to better understand adjustment behavior and to check the parallel trends assumption. The regression model is

$$Y_{iblt} = \sum_{p=1995}^{1997} \delta_p D_{bp} + \sum_{p=1999}^{2004} \delta_p D_{bp} + \sum_{p=1995}^{1997} \theta_p \text{ShareBelow250}_{lp,98} + \sum_{p=1999}^{2004} \theta_p \text{ShareBelow250}_{lp,98} + X_{it}^{'}\gamma + \psi_i + \psi_t + \epsilon_{iblt}, \quad (17)$$

where $\text{ShareBelow250}_{lp,98}$ is the weighted share of firms with at most 250 employees in labor market l and year p. It is zero otherwise. The set of coefficients δ_p still estimates the direct impact of the policy change over time. The set of coefficients θ_p estimates the dynamic effect of agglomeration economies over time. If the effect of the policy on spillovers is causal, then there should not be any differential effect between labor markets before the policy change. I check for this assumption by including coefficients for pre-treatment periods.

4 Data

4.1 Data Sources

The empirical analysis relies on two data sets, the AFID Establishment-Panel by the Federal Statistical Office of Germany and the Establishment History Panel (BHP) by the Institute for Employment Research (IAB).

The AFID dataset has a broad coverage of variables for investment, employment and output for the universe of manufacturing and mining firms with more than 20 employees in Germany. With its unusual richness it perfectly fits the needs for the general empirical analysis. Firm variables are collected through various administrative surveys and are used to inform the government and the public about key economic statistics like aggregate output and investment. Because of the importance of these statistics, firms must provide truthful information by law. The AFID dataset merges the underlying surveys through a unique firm identifier and aggregates information from monthly and quarterly surveys by year. Information is available since 1995 and new waves are continuously added. The dataset is especially suited for the investment analysis since there is separate information for equipment. Further subcategories distinguish different modes of acquisition such as self-production, leasing and purchase. For measures of output, there are revenue, production value, orders and the number of distinct products. Revenue is divided into domestic and foreign. There is however only limited information on labor inputs, with total employment and wage bill being most informative.[24] I also know the industry on a 4-digit level and location on a regional level for each firm.

For a more detailed employment analysis, I use the BHP dataset which provides information on overall employment, employee composition, employee inflows and outflows, and wage. The data are based on the employment histories of the entire labor force covered by social security. They are collected from mandatory communication between firms and the Federal Employment Agency on changes in employment. The BHP aggregates this information on a firm level for 30 June of every year for West Germany since 1975 and for East Germany since 1992. I focus on the years 1995 and 2004. The final sample consists of a 50% random draw of firms and all available years of selected firms are included. Even though individual employee information is lost, employee counts are available for many subcategories. Important categories are level of education, occupation types, full-time employment, vocational training and age. For a better understanding of changes over time, inflow and outflow information provide the number of employees that did not work in the same firm one year before and one year after, respectively. These flows are again divided into subcategories. I use information for incoming and leaving employees that were unemployed one year before and after, respectively. Average wage is based on full-

[24] There is a distinction of employees by contract type. I do not use this information since it does not translate well to other economic concepts.

time employees and available for quartiles and by education. The dataset includes firms of all industries. Information on a 3-digit industry level and a region variable allows the selection of relevant firms.

The AFID and BHP dataset are distinct and it is not possible to link them. Therefore, it is necessary to calculate key policy variables separately and take into account changes in the data collection over time independently. I explain the general adjustments here but delegate further details to Appendix C.

One important variable in the analysis is firm size, since it is the basis for classifying treatment and control group. The program definition requires information on the head count of overall firm employment and vocational trainees.[25] The AFID dataset lacks information on the latter. I match the number of vocational trainees for the years 1999 to 2001 from the cost structural panel (KSE) by the Federal Statistical Office.[26] For observations that are unmatched, I impute values assuming a constant share of trainees within firms or if unknown within industries. For the BHP dataset, on the other hand, information on marginally employed is missing for years before 1999. I impute missing observations by assuming a constant share of marginally employed within firms or if unknown predict the share within industries for different firm size.[27] The change in reporting of marginally employed unfortunately coincides with the policy change. To reduce the risk that the imputation of marginally employed influences the estimation, I drop firms with an average share of more than 25%, which is above the 95th percentile within the manufacturing industry.

Another issue concerns the continuity of firm identifiers. In the AFID data, firm identifiers are constant over changes in firm structure, in the BHP this is not the case. Moreover, in the BHP the firm identifier can change for relatively simple reasons such as changes in the legal form. To have approximately similar definitions of firm identifiers, I adjust them as following. For the AFID dataset, I create new firm identifiers when there are changes in overall firm structure. I also exclude those with more than one-hundred

[25]The definition considers the business year start for calculations. Employment figures in the AFID and BHP dataset do not have this same timing, however, an auxiliary analysis using the IAB establishment survey does not show a systematic difference of employment levels in manufacturing firms within a given year.

[26]The KSE is a yearly firm survey of a stratified random sample in the manufacturing and mining industry and focuses on the production process.

[27]Since I rely on broad firm size intervals in the analysis, and vocational trainees and marginally employed constitute a small share of a firm, measurement error from both imputations should lead to relatively few misclassifications into treatment and control group.

different units in a given year. For the BHP, I follow Hethey-Maier and Schmieder (2013) and create unique identifiers for firms that are connected through employee flows.

Further adjustments are the reclassification of regions as of 2014, the classification of regions into labor markets according to Dustmann and Glitz (2015) and a classification of 2-digit industries that further aggregates uncommon industries.

For the analysis of composition changes by skill, I rely on information of industry-specific ICT usage from several data sources. Sauer and Strobel (2015) provide investment information for 2014 based on data by the Federal Statistical Office and a firm survey by the ifo institute. The EU KLEMS project, in collaboration with the DIW, provides various capital input measures since 1991 for Germany. I rely on the real fixed capital stock of 1998. The IAB establishment survey includes extensive margin investment decisions on firm level for 1993-2014. I aggregate this information on 2-digit industries after controlling for broad regional areas, firm size, average wage and year as possible confounding factors. Finally, the Economic Census collects capital expenditure information for U.S. manufacturing firms. I select year 2002 as the earliest publicly available one.

For the analysis of spillovers, I use the input-output matrix based on domestic and imported production by the Federal Statistical Office for 1998. I use this information to create a treatment intensity measure that considers the degree of interaction between firms of different industries. For this measure, I calculate the weighted share of firms below the cutoff by industry within each labor market. I also calculate input linkages between all industries. I then sum up all shares weighted by the strength of linkage.

4.2 Sample Selection

The sample selection follows the eligibility criteria of the program. First, I select firms in manufacturing industries[28] for years 1995 to 2004 within East Germany excluding Berlin. To avoid peculiar behavior of entering and exiting firms, I also condition on them having economic activity throughout the period of the analysis.

Second, I check for bunching behavior around the cutoff and exclude firms and observations accordingly. Figure 4 displays the size distribution of manufacturing firms in East Germany and West Germany around the cutoff for 1999 to 2004. For West Germany a decrease in the density with increasing firm size is apparent which is as expected (e.g.

[28]These include all industries with a WZ 1993 classification of D.

Axtell, 2001). For East Germany this relation holds generally true as well, however, just below the cutoff there is excess mass. This points to bunching of East German manufacturing firms. I therefore exclude all firms with a size of between 226 and 274 employees in 1998.[29] I further exclude observations within that size interval in any other year.[30]

As a last step, I include only firms with at least 40 employees and a maximum of 1,500 in 1998 and observations that lie within the same interval for any other year. This reduces the problem from biases due to heterogeneous effects among observationally different firms (Heckman et al., 1997).[31] To check whether the choice of the excluded interval and the size interval has an impact on estimation results, I run robustness test that vary these interval boundaries.

Figure 5 considers the impact of the exclusion around the cutoff on firms moving outside their assigned treatment status. As a comparison, I include the case without restriction. The specific sample selection has little impact on the treatment group. At most 3.5% of firms move above the cutoff and once observations around the cutoff are excluded this share reduces to a maximum of 1.5%. There is more movement within the control group. Without the exclusion of firms around the cutoff, 8.6% of firms in the control group fall below the cutoff already in 1999. In 2003 this share is at 26.4%. It is likely that some of these firms move on purpose to take advantage of higher tax credit rates. For the sample with excluded bandwidth, the share of firms moving below is considerably lower. After the policy change the share increases slowly over time, reaches 6.8% in 2001 and is highest in 2004 with 16.9%. Even though the share in 2004 is non-negligible, in this case, a movement below the cutoff is less problematic since these observations are not affected by bunching behavior.

[29] I determine this cutoff by implementing a structural approach that adapts ideas from Garicano et al. (2016). The maximum likelihood estimation leads to a value of 274.6. Implementation details are available upon request.

[30] The exclusion of single observations is not appropriate for estimating the effect on employment since then there would be a selection based on the dependent variable. In this case, I only condition on firm size in 1998.

[31] I implemented a propensity score matching approach between treatment and control group and found qualitatively similar results to those in the main text.

4.3 Descriptive Statistics

Table 1 presents a selection of firm variables from the AFID dataset in Panel A and the BHP dataset in Panel B for the years 1995 to 2004.[32] I show descriptive statistics for all manufacturing firms in West and East Germany (excluding Berlin), and for treatment and control group of the empirical analysis. Overall, firms in West Germany are larger in many respects compared to those in East Germany. The average number of employees in the AFID dataset is 155.19 in West Germany and 89.42 in East Germany. They have the same likelihood of investing in any given year but in West Germany the investment value is larger. These differences in input factors translate to higher revenue with €30.75 million compared to €13.03 million. Panel B reports very similar differences for the number of employees. On top it shows that full-time employees earn considerably more in West Germany. In terms of employee composition, there are actually more employees with college degree, more high-skilled occupations and more vocational trainees in East German manufacturing firms.

Turning to the estimation sample, there are again clear differences in size. This is not surprising given the definition of treatment and control group. The treatment group is similar to the average East German firm when comparing means, but the control group is far larger in every respect.[33] Both groups are more similar for employee composition. For example the share of college graduates is 11.16% in the treatment group compared to 14.82% in the control group.

As a step towards the actual estimation, Figure 6 presents the raw means for treatment and control group over time for different investment and employment outcomes. Encouragingly, in each plot treatment and control group have a similar pre-treatment trend. In the upper left panel, there is a continues decrease in the log of total investment that continues after the policy change in 1999. For subsequent years the investment level stays higher for the treatment group leading to a positive investment response. The decision to invest, shown in the upper right panel, is relatively stable with a value close to 100% and there is little differential movement before or after 1999. I plot log employment using AFID data in the lower left panel and BHP data in the lower right panel. The

[32] I provide descriptive statistics for the complete list of relevant variables for the estimation sample in the Appendix.

[33] This does not mean that there is no overlap. Standard deviations are usually large, especially for investment and employee composition.

evolution in both graphs is quite similar. This is remarkable given that the actual data collection was independent of each other and speaks to the quality of both datasets. Until 1999, employment increases for treatment and control group. Subsequently, employment stays constant or decreases in the control group whereas there is continued growth for the treatment group until 2001. After 2001, employment decreases for the treatment group as well at similar levels as the control group. This pattern indicates a positive employment response to the relative reduction in capital costs.

5 Results

5.1 Investment

As a first outcome I look at firm investment. Table 2 summarizes the average effect for several investment measures. In column (1), I look at the log of total investment and find that the policy change leads to 0.234 log points higher investment among firms below the firm size cutoff. Since the policy change only affected tax credits on equipment investment, column (2) reports the estimate for this subcategory. The estimate is slightly higher with 0.251 and statistically significant. Both outcomes measure only the intensive margin. In column (3), I check for differences in the probability of investing. However, there does not seem to be any response with a point estimate of zero and small standard errors. This is likely the case because of the high rate of firms that invest in any given year independent of treatment. In the literature, one outcome of interest is the investment rate (I_t/K_{t-1}) which combines intensive and extensive margin. I do not observe capital in my datasets. As proxy, I consider investment divided by the average total investment during the period of analysis. I find a positive and statistically significant response of 0.171 for total investment and 0.167 for equipment investment.

Taken together, these results show a positive average investment response from tax credits. In order to compare the magnitude to previous findings in the literature, it is necessary to relate the response to the change in capital costs. I calculate an intensive margin semi-elasticity of total investment with respect to the tax credit rate of 3.304.[34]

[34] I average over the difference in tax credit rate between firms below and above the cutoff from 1999 to 2004 and subtract the difference in tax credit rate from 1995 to 1998. This leads to a change in tax credit rate by 7.083%. The calculation assumes that all investments are modernization investments and covered by tax credits. For cases where this assumption does not hold, the calculated elasticity is a lower

This elasticity is similar to findings by Zwick and Mahon (2017).[35] The elasticities in their study are considered large in comparison to previous findings. They provide evidence that the on average smaller firm size in their sample can explain the difference. The same reasoning could explain the large elasticity in my case as well, since firm size is relatively small.

Table 3 considers a few robustness tests for the investment specification. Column (1) reproduces the results from before. Given that investment is highly volatile, in column (2), I consider a sample where I exclude firms with investment growth in 1997 above the 95th percentile. The response is slightly lower for log of total investment and log of equipment investment but there is no response for the probability of investing. In column (3), I only control for average firm wage, firm fixed effects and year fixed effects. The coefficients are again smaller but lead to qualitatively similar results. Finally, in column (4), I only consider single-establishment firms to eliminate inconsistencies between tax credit eligibility and level of production decisions. I exclude again firms with high volatility in their investments. In this case, the average effect becomes slightly larger. There is stil no response in the extensive margin. Overall, the investment response is similar throughout quite distinct specifications and sample selection.

For a better understanding of investment behavior over time, I look at the dynamic specification. Figure 7 summarizes the coefficients including 95% confidence intervals for the intensive and extensive margin. For both measures of the intensive margin, there is an upturn directly in the year after the policy change. In the subsequent year, investment further increases with coefficients of 0.206 log points for total investment and 0.232 log point for equipment investment. Afterwards, coefficients stabilize around 0.2 log points. There seems to be another increase in 2004. Standard errors are relatively large so that not all coefficients after the policy change are statistically significant. For pre-treatment periods, coefficients are close to zero which suggests that treatment and control group follow the same trend. For the extensive margin, there is not any clear dynamic pattern. Coefficients move around zero for periods before and after the policy change leading to the on average zero result.

bound. The extensive margin semi-elasticity of total investment with respect to the tax credit rate is zero.

[35] They find an intensive margin semi-elasticity of investment of 3.7 and an extensive margin semi-elasticity of 3.8.

5.2 Total Employment

Given the positive response in investment, it is then of interest whether there are changes in the use of labor inputs as well. Table 4 presents the estimation results for the effect of the tax credit rate change on employment within firms. For the regression of column (1), I use employment information from the AFID dataset. I find a positive and statistically significant effect of 0.113 log points on total employment. The response is smaller than for investment but still sizable. In column (2), I consider the same employment measure based on the BHP data. In this case the coefficient is 0.087 which is similar in magnitude and confirms the positive employment response. To check whether the employment effect applies to a large share of employees, in column(3) to (5), I consider regular employees, full-time employees and the intersection of both.[36] The average effect is between 0.076 log points and 0.088 log points which is very close to the effect on total employment.

Figure 8 shows the according dynamic results for total employment and full-time employment. For the case of AFID data, there is a continuous increase in total employment over time. In the first year after the policy change employment is 0.033 log points higher. After three years it reaches 0.099 log points and then levels off. For the case of BHP data, the effect on total employment is more immediate. After one year, employment is 0.077 log points higher. Subsequently there is a small drop, that is followed by a slow increase over time reaching 0.109 log points in 2004. When considering only full-time employees, there is again an immediate response of 0.055 log points. Even though there is again a drop in the subsequent year, there is a stronger increase in the effect over time, reaching 0.123 log points in 2004. When checking for parallel trends in employment for pre-treatment periods, coefficients are again close to zero and mostly statistically insignificant. For total employment with the AFID data there seems to be some movement already in 1998, however, it is small in magnitude.

One concern for the employment estimations is the slightly different sample selection of observations around the cutoff, and at the lower and upper bound. Since for employment I only condition on firm size in 1998, there is the possibility that firms bunch around the cutoff in later years and bias the estimation. For this reason, I look at the yearly change in the log of total employment over time in Figure **??** using BHP data. In this case,

[36] These measures largely exclude marginally employed and thus allow to check the influence of the change in reporting of marginally employed in 1999 by comparing it to total employment.

I follow the standard sample selection excluding firms by size in 1998 and observations according to size in any other year, which eliminates bunching observations. In years before the policy change, the effect is close to zero and statistically insignificant. In 1999, there is positive growth in total employment of 0.079. In subsequent years coefficients are statistically insignificant again. They are still in line with the level results suggesting a decrease in employment in 2000 and small positive growth over time in subsequent years.

5.3 Revenue

Because of a higher tax credit rate, firms increase investments and employment. It seems natural to investigate whether at least some of this effect turns into higher output. Table 5 provides estimation results for various revenue measures and the number of distinct products as proxy for output.[37] In column (1), I look at total revenue and find a small effect of 0.011 log points. However, for domestic revenue in column (2), the effect is 0.083 log points and for domestic revenue of manufacturing goods in column (3), it is 0.080 log points. This is puzzling since there must be a reduction in exporting that equalizes changes in domestic production, even though exporting makes up a small share of production on average. I therefore consider the possibility that the results for total revenue are driven by outliers among exporting firms. In column (4), I restrict the sample to observations with an exporting share of maximum 15%. In this case, the effect is 0.076 which is close to the effect on domestic revenue (though statistically insignificant). Thus, the discrepancy between the effect on total revenue and domestic revenue seems to be confined to those firms that already export sizable amounts. As additional outcome I look at the number of distinct products in column (5) but I do not find an effect of tax credits on the number of products.[38]

5.4 Robustness Test

Taken together, these first results provide a positive assessment for investment tax credits. Not only do they increase firm investment, but they also lead to more employment and thereby to higher revenue. The results are all based on a specific selection of the firm size sample. I check how stable these results are when varying on the one hand the interval

[37] The revenue measures mask the actual output response by possible changes in output prices.

[38] I cannot check whether firms substitute products without changing their overall number.

of excluded firms around the cutoff and on the other hand the interval of smallest and largest included firms. Furthermore, the estimation relies on the difference in tax credit rate between firms below and above the cutoff as the defining margin. I test this behavior by looking at several policy irrelevant cutoffs within the treatment group which should lead to zero results.

Table 6 reports coefficients for the most relevant measures of investment, labor and revenue. The interval of excluded firms around the cutoff varies in each column. In column (1), I do not exclude any firm. Column (3) follows the main specification with an excluded interval of 225 to 275 employees. In addition, I report an intermediate case in column (2) and larger intervals in column (4) and (5). Reassuringly, estimates for all outcomes are consistent throughout the different intervals. This applies especially to log overall investment, the extensive margin investment decision and employment. For log equipment investment there are slight differences although there is no clear pattern by the size of the excluded interval. For log domestic production, coefficients grow somewhat by the size of the excluded interval, but qualitative results are the same. Taken together, there is little evidence that the exclusion of firms around the cutoff has a notable impact, possibly because the number of firms that bunch at the cutoff is small in comparison to the whole sample.

Table 7 keeps the excluded bandwidth constant but instead varies the smallest and largest included firm size. In this table, column (4) reproduces the main specification and columns to the left look at smaller intervals whereas columns to the right look at larger intervals. Again, independent of the chosen interval, estimates lead to qualitatively equal results with limited fluctuations in the actual coefficient. The largest differences apply to the investment variables for which coefficients in column (1) are somewhat smaller and those in column (6) larger than the rest. It should be noted that for the smallest firm interval, the number of observations is considerably smaller which may lead to higher susceptibility to outliers.

Finally, Table 8 looks at the result for placebo cutoffs. The sample only consists of those in the treatment group. I focus on the cutoffs 80, 100 and 125 to have a reasonable number of firms below and above these new cutoffs. I still exclude firms around the cutoffs as in the main specification. For the majority of outcomes, coefficients are close to zero and statistically insignificant. This holds true for all placebo cutoffs. For the

extensive margin investment decision, coefficients are negative and statistically significant. However, given the high probability of firms investing in any given year, the coefficient is not economically significant.

5.5 Flow Information

In the robustness tests, coefficients are stable over different sample selections and the irrelevant firm size cutoffs lead to expected zero results. I therefore continue with an in depth analysis of the adjustment process of labor inputs. As a first step, I examine employee flow information. The BHP data reports the number of employees within each firm that did not work there one year prior or that do not work there one year after. For my empirical analysis, I first relate the inflow and outflow information to overall employment one year prior. This is equal to the yearly growth rate for net flows and the hypothetical growth rate for inflows and outflows assuming the other flow value to be zero. I then cumulate these rates over time and take logs to have a measure which is similar to log employment.[39] I apply the same procedure to flow information of employees within firms that were unemployed one year prior or are unemployed one year after.

Table 9 reports the results for each of the cumulated flow variables. In column (1) to (3), the dependent variables include flows from all employees. The result for net flows in column (1) is nearly identical to the result using log employment with an increase of cumulated net flows by 0.086 log points. This serves as a check on the validity of the analysis of flow information. In column (2), I focus on inflows and find an increase by 0.102 log points. Since the additional inflows in treated firms are larger than net flows, there must be an increase in outflows as well. This is confirmed in column (3), although the coefficient is small and statistically insignificant. These results show that firms that received relatively more tax credits after the policy change increased firm size predominantly through hiring additional employees. Importantly, the point estimate does not suggests that the control group has higher outflows which counters the concern that the employment effect is due to a shift of employees from the control group to the treatment group. If anything, the higher cumulated outflow among treated firms suggests changes in employment structure.

From a policy perspective, it is of interest whether unemployed individuals gain from

[39] This is the case since firm fixed effects control for average firm employment.

tax policies. I analyze the flows related to unemployment in column (4) to (6). Even though the coefficients are statistically insignificant, their magnitudes are economically significant. The effect of inflows from unemployment in column (5) is 0.050 log points. When comparing this estimate to the one for overall inflows, the hiring of unemployed people constitutes 49% among the additionally hired employees. This is sizable and similar to the share of all hirings of 60%. The effect on the cumulated outflow rate into unemployment is 0.018 log points which is almost as high as for overall outflows. Thus, it seems that the additional separations are of employees who have problems finding re-employment.

I consider the dynamic effects of the flow variables in Figure 10. Taken together, the dynamic results confirm the previous findings. It is of interest that the net flow and inflow for unemployment are statistically significant in this specification. It is also the case that the share of inflows from unemployment is larger at the beginning with 71.4% and reduces over time. Furthermore, the increase in outflows, although still statistically insignificant, slowly increases over time which suggest that firms first hire additional employees and only over time let go of unnecessary employees.

5.6 Capital-Skill Complementarity

The analysis of flow information provides suggestive evidence that there are changes in employment structure. In the literature, there is an important discussion about capital-skill complementarity. I examine this adjustment mechanism in the context of tax credits by looking at firm ratios of high-skilled labor, high-skilled occupations and manual occupations. I classify the skill level of labor according to education and divide between college educated and others. I build the same ratio based on full-time employees to see whether there are differences for example because of marginally employed. For the classification of different occupations, I build on the pre-defined Blossfeld classification. I consider technicians, engineers, semi professions, professions and managers as high-skilled occupations. As another dimension, I am interested in the shift of manual labor. These are occupations that are either low-skilled or medium-skilled. I compare them to low-skilled and medium-skilled service and sales occupations. Finally, within manual labor, I compare medium-skilled (qualified) to low-skilled occupations.

In Table 10, I look at the average effect of each ratio. They are all close to zero

and statistically insignificant. Given that the estimate is in log points of a ratio, I can reject large changes in employment composition. For example, for the ratio of college educated the upper bound of the 95% confidence interval is 0.052 log points which would shift the high-skilled labor ratio just from 0.131 to 0.138. This zero result is surprising given the large literature concerning aggregate changes in inequality and job polarization. This literature considers ITC as important mechanism for these shifts which started in the 1990s. It is possible that in my setting, ITC does not play a big enough role and therefore does not influence the average employment composition. This however does not mean that there is no influence more generally. To explore this issue in more detail, I look at heterogeneous effects of tax credits by the intensity of ITC usage. Since the main datasets do not have information on this type of capital, I match various data on an industry level.

Table 11 presents the estimation results using various measures on ICT. I find that firms in industries with higher ICT shares change their employment composition more towards high-skilled labor and high-skilled occupations when receiving investment tax credits. This heterogeneous response is remarkably stable for each ICT measure, although it is not always statistically significant, even though they are from quite distinct sources. The effect on manual occupations and on shifts within manual occupations is noisy. Depending on the measure for ICT, the response is either negative or positive. Nevertheless, using information from the U.S. Economic Census, I find a statistically significant effect for a larger shift towards qualified manual labor among industries with higher ICT shares.

5.7 Spillover Effects

All results so far speak to the direct effect of investment tax credits. I now exploit the regional variation in the number of firms below the cutoff and thus variation in the number of firms that benefited more from the policy change to examine spillover effects.

I start with an analysis that does not consider differences in treatment intensity through linkages between industries. Table 12 reports the results according to specification 16. I also show modifications that vary the type of fixed effects included in the regression and the firm size interval. The table includes estimates for the direct effect and the spillover effect. I find a consistent direct effect of tax credits on employment. For

the main specification in panel 1 column (1), the coefficient is 0.081 which is similar to the previous regressions. For all other specifications the direct effect is quite similar. The indirect effect is not as robust. In the main specification, the coefficient is 0.272, statistically significant and stable when including industry-year fixed effects and area-year fixed effects. However, when increasing the firm size interval, the magnitude of the coefficient drops considerably. With a lower firm size boundary of 10 employees, the coefficient is only 0.054.

To better understand the differences in these coefficients, I plot the dynamic effect for the spillover analysis in Figure 11. There are visible pre-treatment trends for specifications with a smaller firm size interval. However, for the interval with a lower bound of 20, coefficients are close to zero in pre-treatment periods. The effect picks up directly after the policy change and increases further over time. This suggest that there are important spillover effects from the tax credit policy.

It is a common view, that the magnitude of spillovers varies within a region depending on how benefits from one firm propagate to others. One propagation mechanism considers input and output linkages between industries. I examine this mechanism by varying the treatment intensity not only on labor market level but also on an industry level using actual input flows between industries as weighting factor. Table 13 reports the results for the main specification in column (1) and for increasing firm size intervals in columns (2) to (4). The coefficients for the direct effect stay stable throughout all specifications again. The coefficients for the spillover term are relatively stable as well, fluctuating between 0.128 and 0.196, pointing again to the existence of spillovers. Figure 12 plots the according dynamic effects.

6 Conclusion

In this paper, I empirically assess a tax credit program in Germany to estimate the causal impact of a capital cost reduction on firm input choices. To evaluate the success of this program, I go beyond a firm investment analysis and look at the effect on employment of different labor types and the influence of spillovers.

I find that firms increase both their investment and employment substantially and that this then translates to higher revenue. For employment, the effect occurs through

hiring additional employees, including a sizable share of unemployed. I do not find a shift in the employment composition by skill types. Nevertheless, industries with higher dependence on ICT technology are more likely to shift to high-skilled labor. Finally, spillovers further increase the employment effect.

These results are encouraging for the use of investment incentive programs in fiscal policies. The fact that there is a benefit for unemployed and low-skilled individuals is important from a welfare perspective since it is believed that they have more to gain. However, more research is needed to assess whether investment incentives are the most efficient way of improving economic outcomes and how programs directly targeted to employees compare during short-term and long-term.

Whether the results will transfer to other settings is open for debate. Since tax credits reduce capital costs independent of firm profits they should be appropriate throughout the business cycle and effective in economic downturns. However, firms in the German tax credit program were particularly hard hit and had much more potential to catch up.

The result on the larger shift towards high-skilled among high share ICT industries should be a warning sign even though I do not find changes on average. Given that nowadays ICT becomes more important in the production process and that robots already take on many simple tasks, incentivizing investments in the future may lead to less beneficial outcomes for unemployed and low-skilled individuals.

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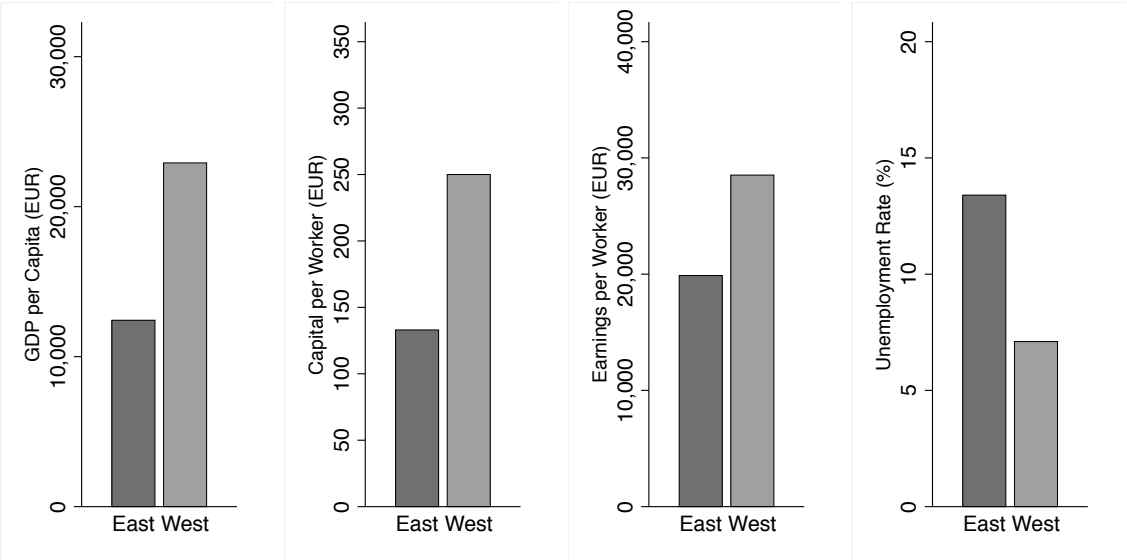


Figure 1: Economic Indicators for West and East Germany for 1991-1994

Note: The bars represent averages over the years 1991-1994. Data source for Panel (a)-(c) is the Federal Statistical Office. Data source for Panel (d) is the Federal Employment Institute. The unemployment rate is based on figures of June of each year. All of Berlin is counted towards the statistics of East Germany.

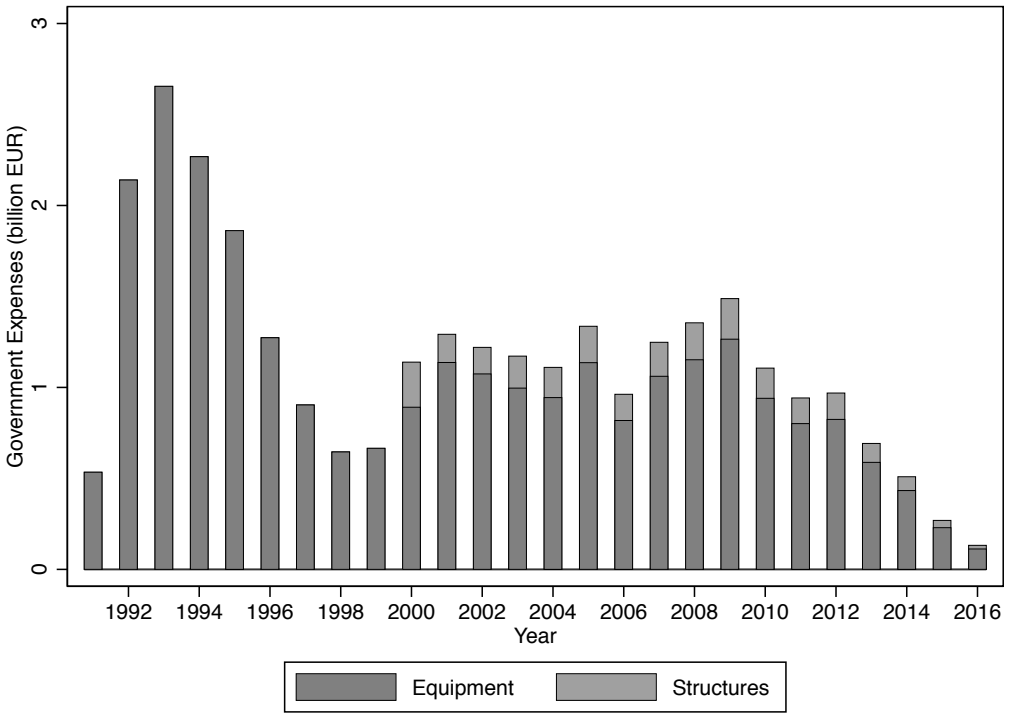


Figure 2: Government Expenses for Tax Credits by Year and Investment Type

Note: Information taken from subsidy reports of the German government. Values that are available only in Deutsche Mark were converted to euro using the official conversion rate. Figures on government expenses get revised over time due to additional information. The presented figures are taken from the most recent report available for each year respectively to reflect the most current information status. Expenses after 2016 excluded.

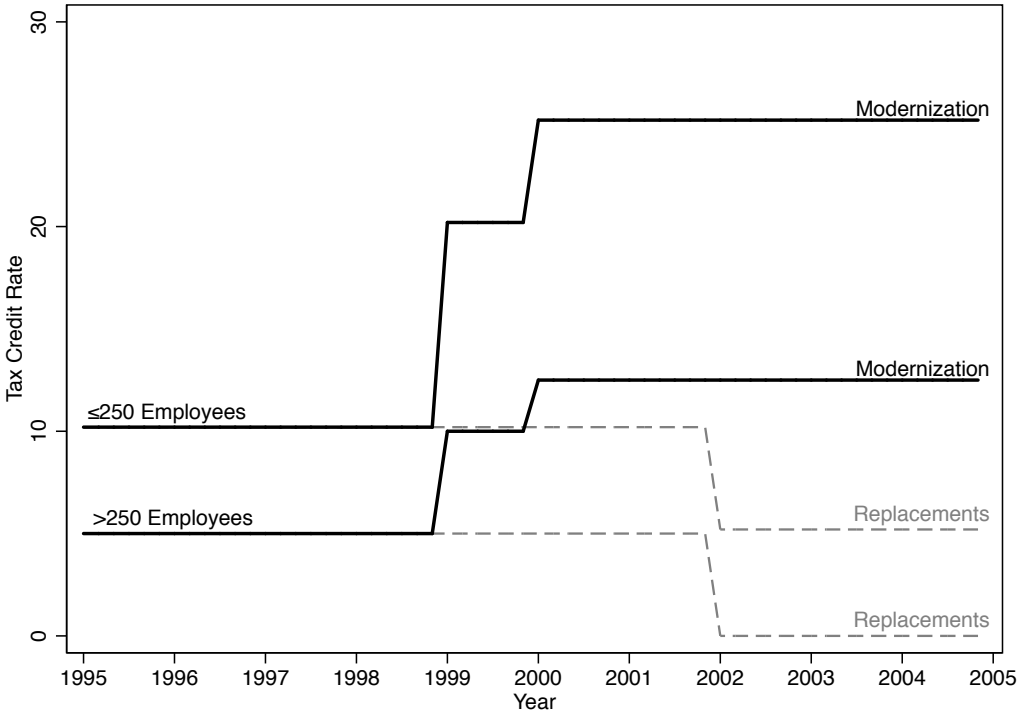


Figure 3: Equipment Tax Credit Rates for Manufacturing Firms

Note: The shown tax credit rates apply to manufacturing firms in most parts of East Germany excluding Berlin for equipment investment that start and end at the same day. For Berlin and in some years for areas close to Berlin rates were lower. There was a slight increase in the tax credit rate for modernization investments in regions close to the Polish or Czech border starting in 2001. Changes in tax credit rate were usually accompanied by phase-out periods to allow a constant rate for longer-lasting investment processes.

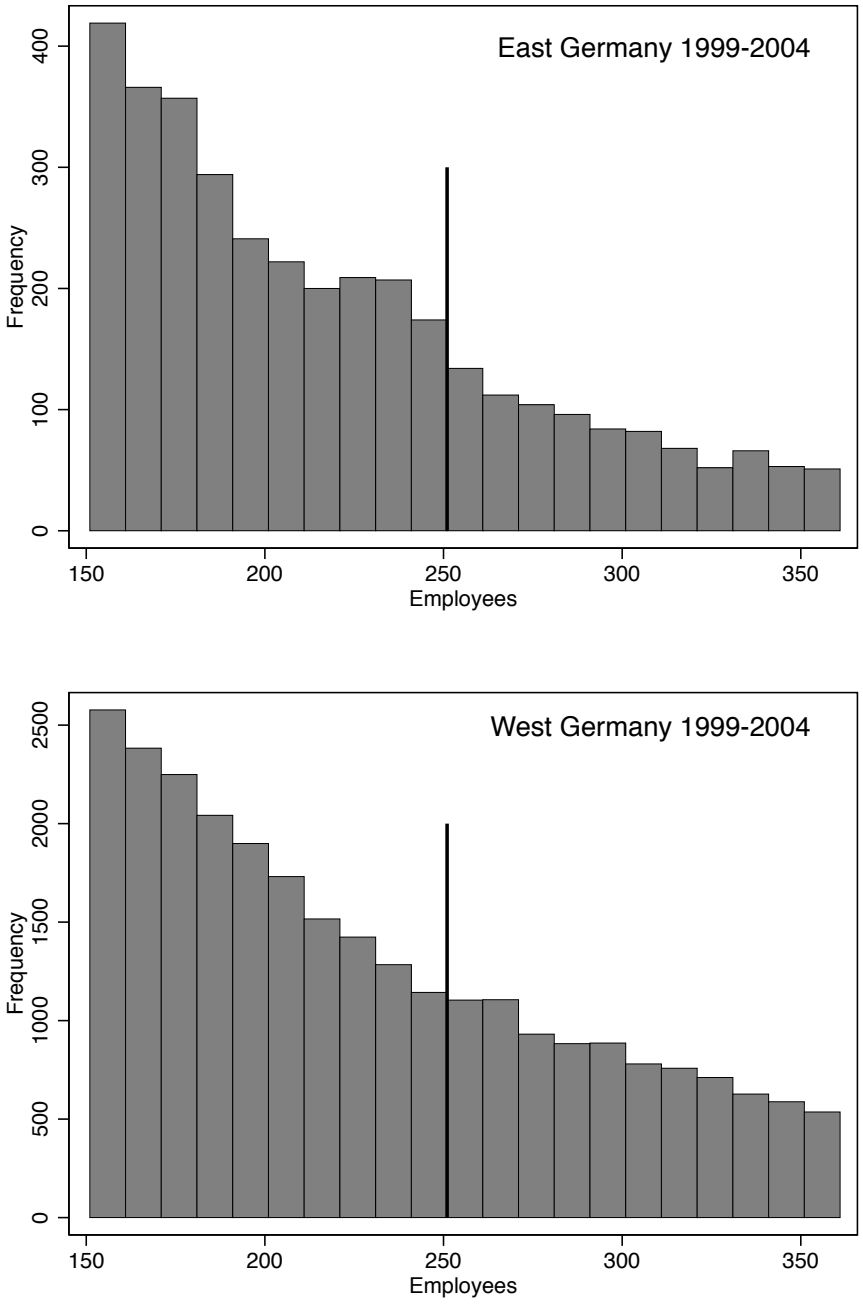


Figure 4: Firm Size Distribution of Manufacturing Industry

Note. Data from AFID establishment panel. The sample consist of all observations of manufacturing firms for East and West Germany excluding Berlin between 1999 and 2004. Firm size is based on total number of all employees but excludes vocational trainees.

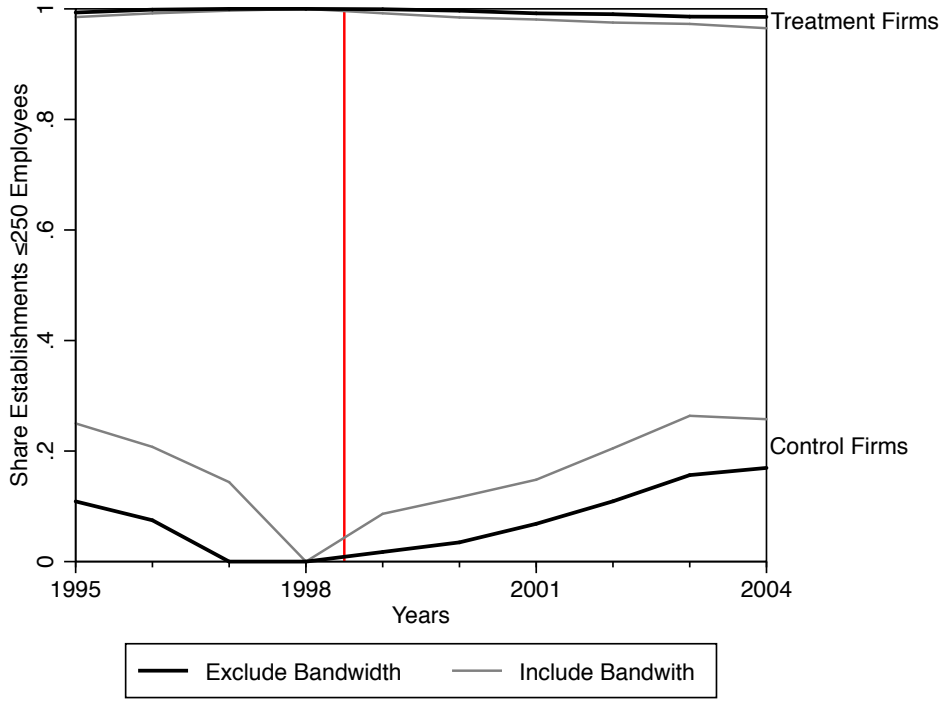


Figure 5: Movement of Treatment and Control Group Around Firm Size Cutoff

Note: Data from AFID establishment panel. The figure shows the share of firms in the treatment and control group below the firm size cutoff. Firm size is defined as total employee head count minus vocational trainees. The treatment group consists of firms with 250 or fewer employees. The sample selection for 'Exclude Bandwidth' is according to main text and specifies the exclusion of firms and observations around the firm size cutoff. The case of 'Include Bandwidth' follows the same sample selection, but keeps firms and observations around the size cutoff.

Table 1: Descriptive Statistics for Manufacturing Firms

Panel A. AFID Data	West Germany	East Germany		
	All	All	Treatment	Control
No. of Employees (mean)	155.19	89.42	91.88	508.19
No. of Employees (median)	57	47	75	414
Investing (%)	86.0	86.0	94.0	97.0
Investments (thousand EUR)	1,267.50	1,172.35	805.96	8,605.11
Revenue (million EUR)	30.75	13.03	11.39	101.73
Part of Multi-Establishment Firm	22.0	22.0	9.98	15.76
Observations	357,662	64,154	16,374	1,266

Panel B. BHP Data	West Germany	East Germany		
	All	All	Treatment	Control
No. of Employees (mean)	129.24	81.09	90.00	470.52
No. of Employees (median)	47	41	72	406
Full-time employees (mean)	111.63	70.95	80.73	425.48
Average Daily Wage Full-time	85.86	58.16	59.49	78.53
Share College Degree (%)	6.43	10.58	11.16	14.82
Share High-Skilled Occupation (%)	12.63	13.06	13.02	17.00
Share Vocational Trainees (%)	3.89	5.30	4.34	4.16
Observations	212,924	37,239	9,126	990

Note: Statistics are based on firms in the manufacturing sector for the years 1995-2004 excluding Berlin. The number of observations is based on the according statistic for number of employees. The group of all manufacturing firms includes those with more than 20 employees to allow for a better comparison between AFID and BHP data. Treatment and control group are according to the estimation sample.

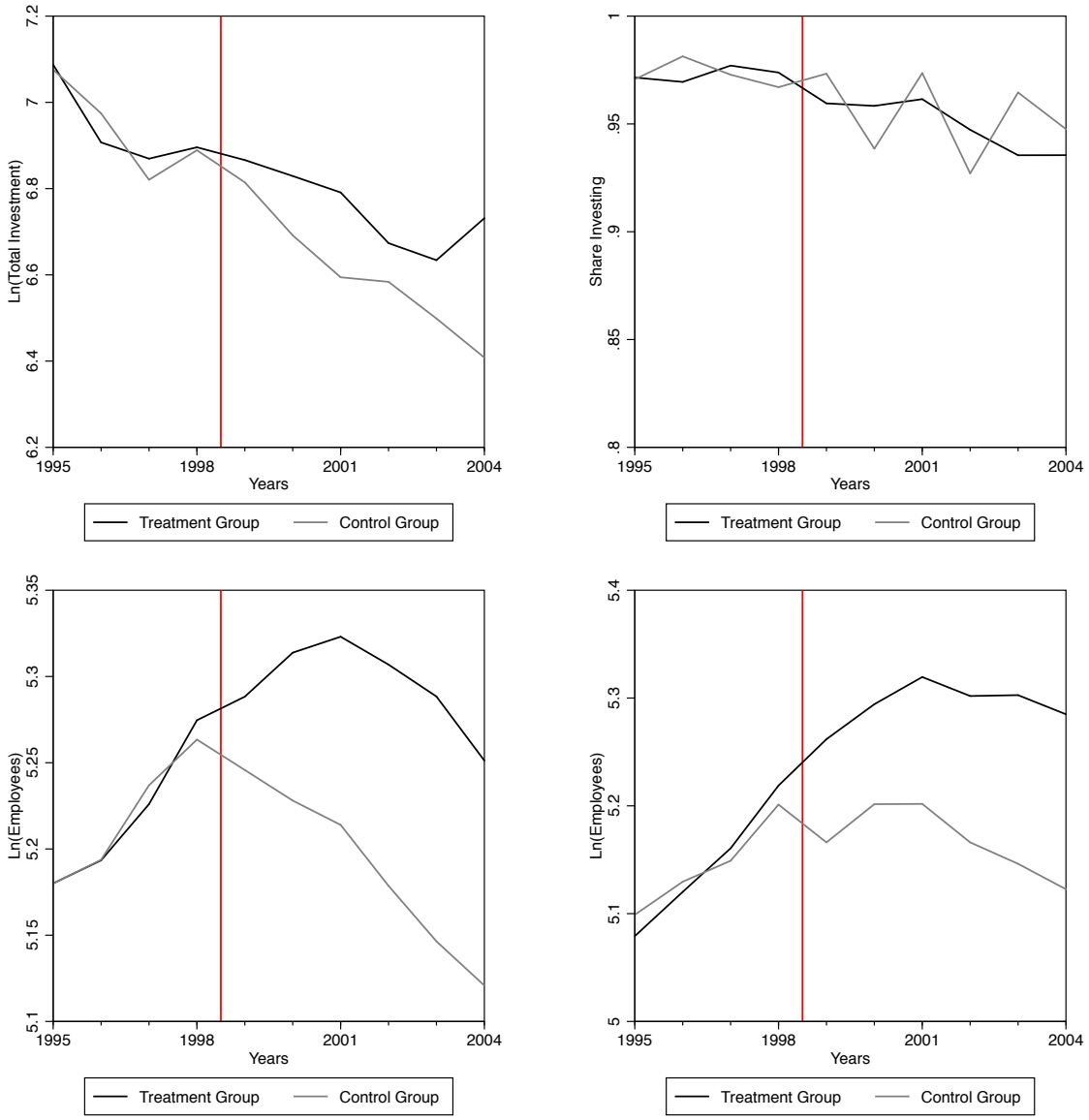


Figure 6: Average Firm Investment and Employment By Treatment Status

Note. The figures report raw means for each outcome by treatment and control group over the period 1995 to 2004. Outcome variables are the log of total investment in the upper left panel, share of firms with positive total investment in the upper right panel, log of employment based on AFID data in the lower left panel and log of employment based on BHP data in lower right panel. For each data point, I subtract the group mean for the pre-treatment period (1995-1998) and add the pooled mean for the same period for facilitating a comparison of trends.

Table 2: Difference-in-Differences - Investment

	Dependent Variable:				
	$\log\left(\begin{matrix}\text{Total}\\\text{Investment}\end{matrix}\right)$	$\log\left(\begin{matrix}\text{Equipment}\\\text{Investment}\end{matrix}\right)$	Investing (1/0)	$\frac{\text{Total Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Equip Invest}_t}{\text{Avg Tot Invest}}$
	(1)	(2)	(3)	(4)	(5)
Treated	0.234**	0.251***	-0.000	0.171**	0.167**
	(0.089)	(0.092)	(0.008)	(0.081)	(0.068)
Observations	15,275	15,071	15,900	15,865	15,865

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total investment in column (1), log of equipment investment in column (2), a dummy for having positive total investment in column (3), total investment over the average total investment between 1995-2004 in column (4) and equipment investment over the average total investment between 1995-2004 in column (5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 3: Difference-in-Differences - Specification Robustness Investment

	(1)	(2)	(3)	(4)
	Dependent Variable: Log(Total Investment)			
Treated	0.234**	0.218**	0.201**	0.270***
	(0.089)	(0.090)	(0.090)	(0.094)
Observations	15,275	14,547	15,275	13,047
	Dependent Variable: Log(Equipment Investment)			
Treated	0.251***	0.231**	0.227**	0.285***
	(0.092)	(0.093)	(0.092)	(0.099)
Observations	15,071	14,347	15,071	12,880
	Dependent Variable: Investing (1/0)			
Treated	-0.000	-0.000	-0.004	-0.002
	(0.008)	(0.008)	(0.007)	(0.009)
Observations	15,900	15,148	15,900	13,555

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total investment in the upper panel and the log of equipment investment in the lower panel. Column (1) reproduces the coefficient from Table 2. Column (2) excludes firms with volatile investment measured by growth of total investment in 1997 above the 95th percentile, column (3) only includes log average firm wage, firm fixed effects and year fixed effects as control and column (4) is conditional on a a sample of single-establishment firms and excluding firms with volatile investment measured by growth of total investment in 1997 above the 95th percentile. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

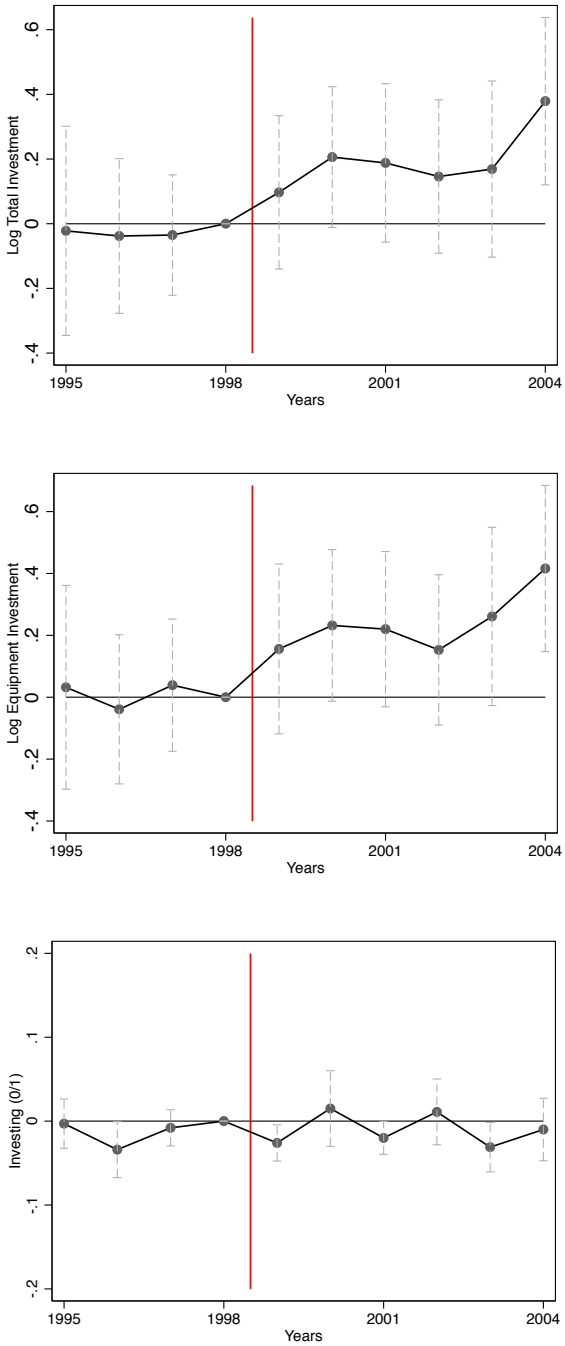


Figure 7: Dynamic Effect of Policy Change on Investment Behavior

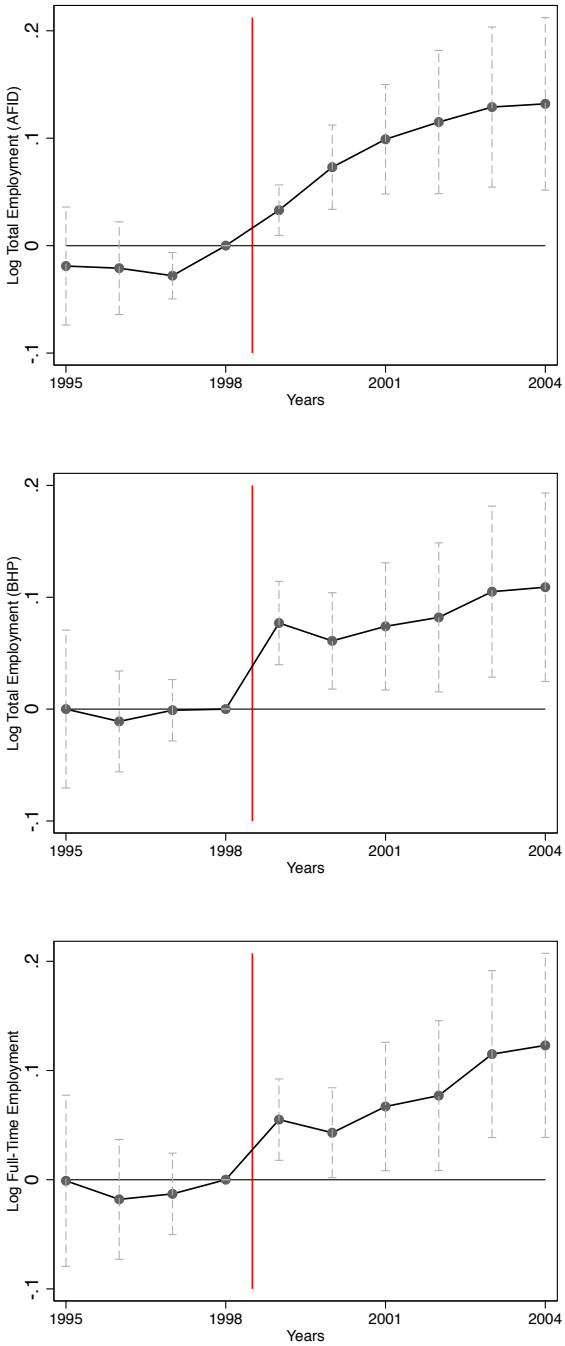
Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total investment in the upper panel, log of equipment investment in the middle panel and a dummy for having positive total investment in the lower panel. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Firms with volatile investment, measured by growth of total investment in 1997 above the 95th percentile, are excluded. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 4: Difference-in-Differences - Employment

	Dependent Variable: Log Employment of				
	Total (AFID)	Total (BHP)	Regular	Full-Time	Full-Time-Regular
	(1)	(2)	(3)	(4)	(5)
Treated	0.113***	0.087**	0.076**	0.088**	0.087**
	(0.031)	(0.037)	(0.036)	(0.038)	(0.038)
Observations	17,637	10,116	10,116	10,116	10,116

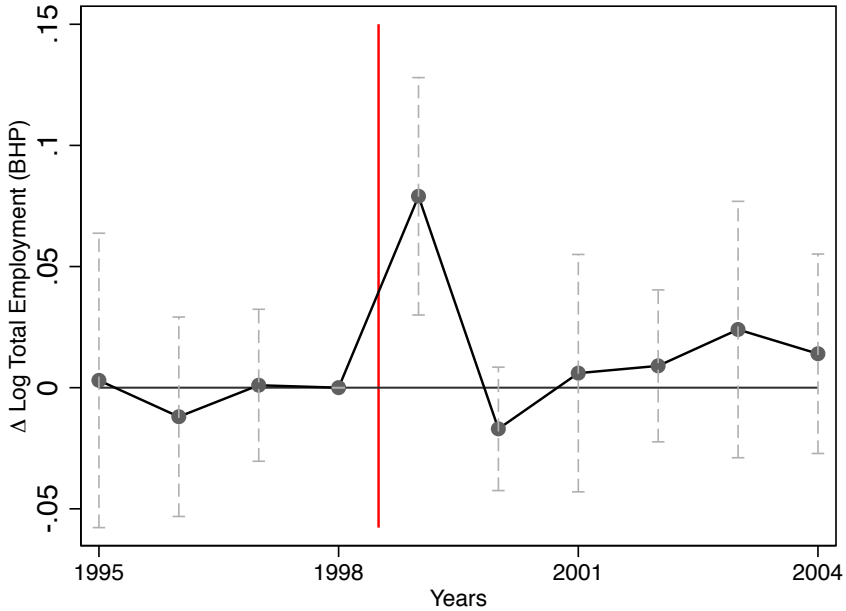
Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total employment in column (1) and column (2), log of regular employees in column (3), log of full-time employees in column (4) and log of regular full-time employees in column (5). Data from AFID in column (1) and BHP in column (2)-(5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Figure 8: Dynamic Effect of Policy Change on Firm Employment



Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Figure 9: Dynamic Effect of Policy Change on Firm Employment Growth



Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 5: Difference-in-Differences - Effect on Output

	Dependent Variable:				
	Log Revenue (1)	Log Domestic Revenue (2)	Log Domestic Manuf Revenue (3)	Log Revenue (low exporting) (4)	Log Products (5)
β	0.011	0.083**	0.080*	0.076	0.022
	(0.037)	(0.038)	(0.040)	(0.050)	(0.032)
Obs	15,906	15,898	15,897	11,689	15,547

Note. Each coefficient is estimated from different regression following Specification 14. The dependent variables are log of total revenue in column (1), log of domestic revenue in column (2), log of domestic revenue for manufacturing output in column (3), log of revenue in column (4) and log of the number of distinct products in column (5). The sample in column (4) is conditional on an exporting share of below 0.15. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 6: Impact of Excluded Bandwidth Around Cutoff

	Excluded Bandwidth:				
	None	[238,262]	[225,275]	[213,287]	[200,300]
	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: Log(Total Investment)				
Treated	0.209**	0.208**	0.218**	0.212**	0.210**
	(0.082)	(0.085)	(0.090)	(0.096)	(0.098)
Observations	15,559	15,014	14,547	14,168	13,862
	Dependent Variable: Log(Machinery Investment)				
Treated	0.207**	0.189**	0.231**	0.218**	0.215**
	(0.087)	(0.090)	(0.093)	(0.102)	(0.103)
Observations	15,354	14,813	14,347	13970	13,674
	Dependent Variable: Investing (0/1)				
Treated	-0.000	-0.004	-0.000	-0.002	-0.001
	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
Observations	16,186	15,625	15,071	14,763	14,446
	Dependent Variable: Log(Employees) AFID				
Treated	0.091***	0.105***	0.113***	0.108***	0.109***
	(0.030)	(0.031)	(0.031)	(0.033)	(0.034)
Observations	18,377	18,007	17,637	17,287	16,997
	Dependent Variable: Log(Employees) BHP				
Treated	0.080**	0.080**	0.087**	0.107**	0.114**
	(0.036)	(0.036)	(0.037)	(0.044)	(0.046)
Observations	10,406	10,256	10,116	9,876	9,706
	Dependent Variable: Log(Domestic Revenue)				
Treated	0.058*	0.074*	0.083**	0.088**	0.089**
	(0.032)	(0.038)	(0.038)	(0.040)	(0.039)
Observations	16,988	16,382	15,898	15,510	15,191

Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following main specification (14). The excluded firm and observations (for employment only firm) are according to the column titles. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 7: Impact of Firm Size Interval

	Firm Size Interval					
	[80,225], [275,750] (1)	[54,225], [275,1125] (2)	[45,225], [275,1350] (3)	[40,225], [275,1500] (4)	[34,225], [275,1800] (5)	[20,225], [275,3000] (6)
	Dependent Variable: Log(Total Investment)					
Treated	0.165*	0.244***	0.211**	0.218**	0.284***	0.327***
	(0.093)	(0.091)	(0.091)	(0.090)	(0.089)	(0.096)
Observations	6,307	10,459	12,821	14,547	17,043	23,945
	Dependent Variable: Log(Machinery Investment)					
Treated	0.163*	0.262***	0.224**	0.231**	0.273***	0.300***
	(0.092)	(0.093)	(0.092)	(0.093)	(0.092)	(0.095)
Observations	6,218	10,332	12,657	14,347	16,799	23,566
	Dependent Variable: Investing (0/1)					
Treated	0.008	0.013	0.003	-0.000	-0.003	-0.020***
	(0.010)	(0.010)	(0.008)	(0.008)	(0.008)	(0.007)
Observations	6,464	15,625	13,308	15,148	17,819	25,494
	Dependent Variable: Log(Employees) AFID					
Treated	0.091**	0.101***	0.105***	0.113***	0.135***	0.128***
	(0.039)	(0.033)	(0.032)	(0.031)	(0.034)	(0.033)
Observations	7,964	12,968	15,626	17,637	20,455	27,717
	Dependent Variable: Log(Employees) BHP					
Treated	0.097**	0.097**	0.089**	0.087**	0.101***	0.115***
	(0.045)	(0.040)	(0.037)	(0.037)	(0.035)	(0.034)
Observations	4,540	7,406	9,086	10,116	11,826	19,244
	Dependent Variable: Log(Domestic Revenue)					
Treated	0.090**	0.096**	0.082**	0.083**	0.111***	0.121***
	(0.045)	(0.041)	(0.040)	(0.038)	(0.041)	(0.042)
Observations	6,731	11363	13,987	15,898	18,701	26,796

Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following main specification (14). The sample consists of firms according to the size intervals given in the column titles. Except for log employment these intervals apply to observations as well. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 8: Placebo Cutoffs Within Treatment Group

	Dependent Variable				
	Log Total Investment	Log Machinery Investment	Investing (1/0)	Log Employees (AFID)	Log Employees (BHP)
	(1)	(2)	(3)	(4)	(5)
Cutoff: 80					
Treated	-0.043	-0.038	-0.030*	0.033	0.006
	(0.078)	(0.076)	(0.016)	(0.026)	(0.030)
Observations	6,552	6,448	6,803	9,939	5,410
Cutoff: 100					
Treated	-0.027	0.023	-0.024**	0.039	0.009
	(0.068)	(0.072)	(0.012)	(0.025)	(0.027)
Observations	8,623	8,510	9,011	12,237	7,036
Cutoff: 125					
Treated	0.010	0.040	-0.020*	0.020	0.026
	(0.079)	(0.076)	(0.011)	(0.027)	(0.036)
Observations	10,385	10,247	10,886	14,025	7,916

Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following a modified version of specification (14). I change the firm size interval to between 50 and 250 employees. I set a cutoff for treatment and control group to 80 in the first panel, 100 in the second panel and 125 in the third panel. I exclude firms in an interval between -24 and +24 of the cutoff. Except for log employment these intervals apply to observations as well. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 9: Difference-in-Differences - Analysis of Flow Data

	Dependent Variable:					
	Flows All Employees			Flows Unemployment		
	Net	In	Out	Net	In	Out
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.086**	0.102**	0.022	0.031	0.050	0.018
	(0.037)	(0.041)	(0.032)	(0.019)	(0.034)	(0.026)
Observations	10,099	10,099	10,099	10,099	10,099	10,099

Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of cumulated net flows growth in column (1), the log of cumulated inflow growth in column (2), the log of cumulated outflow growth in column (3), the log of cumulated net flow growth from or to unemployment in column (4), the log of cumulated inflow growth from or to unemployment in column (5) and the log of cumulated outflow growth from or to unemployment in column (6). Growth rate is defined as a flow over past year total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

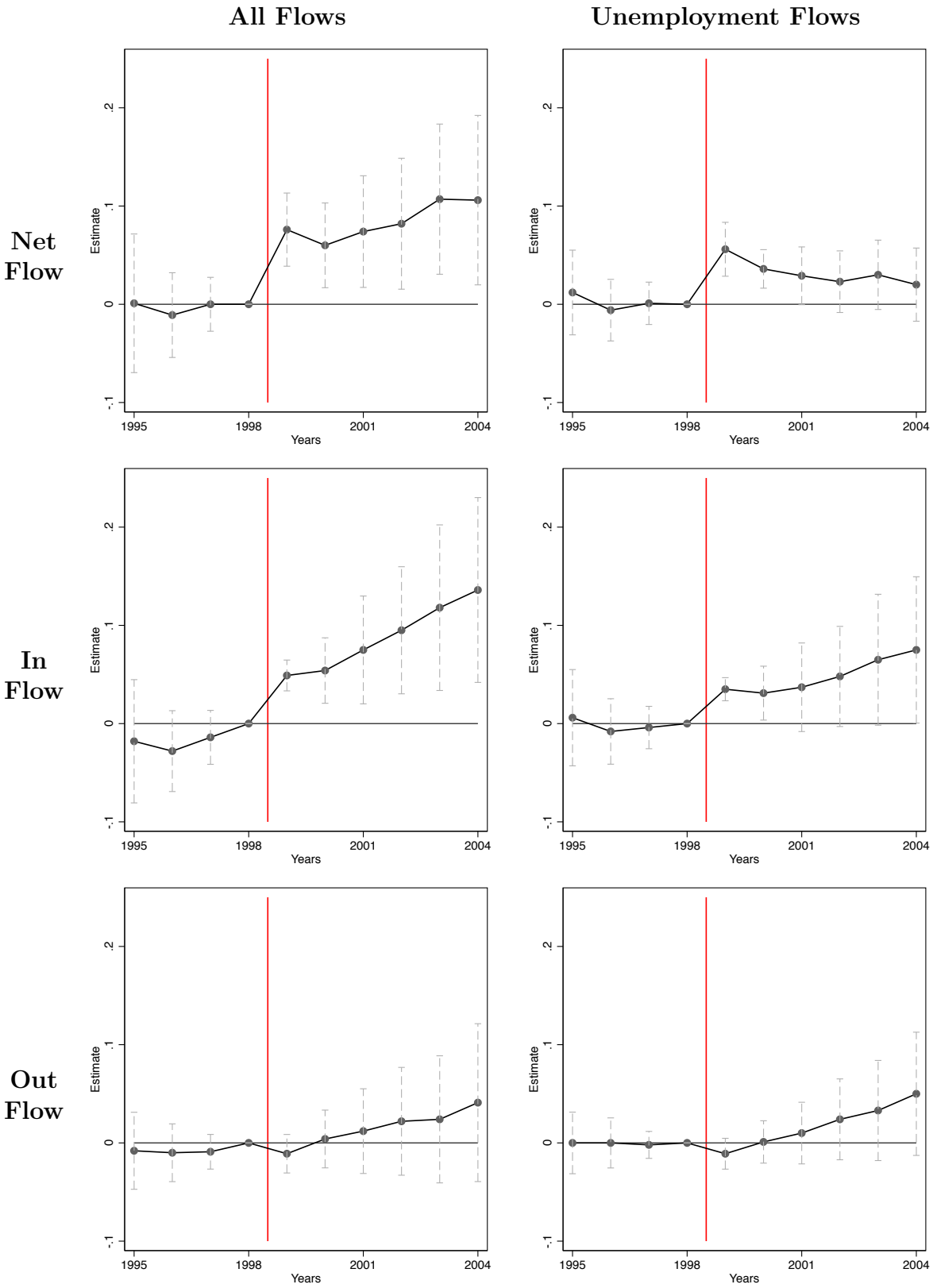


Figure 10: Dynamic Effect By Flow Type

Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of cumulated flow growth. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 10: Differences-in-Differences - Skill Ratios

	Dependent Variable: Log Ratio of				
	College Educated	Full-time College Educ	High-Skilled	Manual Labor	Qualified Within Manual Labor
	(1)	(2)	(3)	(4)	(5)
Treated	-0.011	0.014	-0.027	-0.019	0.004
	(0.032)	(0.031)	(0.043)	(0.045)	(0.043)
Average Share (%)	11.6	12.3	13.4	74.2	42.2
Obs	8,521	8,466	8,623	8,968	8,008

Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of college educated vs. non-college educated in column (1), the log of full-time college educated vs. full-time non-college educated in column (2), the log of high-skilled occupations vs. low-skilled occupations in column (3), the log of manual occupations vs. service occupations in column (4) and the log of qualified manual occupations vs. unqualified manual occupations in column (5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * $p<0.10$, ** $p<0.05$, *** $p<0.01$

Table 11: Differences-in-Differences - Employment Shares By Skill Level

	Dependent Variable: Log Ratio of				
	College Educated	Full-time College Educ	High-Skilled	Manual Labor	Qualified Within Manual Labor
	(1)	(2)	(3)	(4)	(5)
Treated	-0.021	0.005	-0.041	-0.009	-0.006
	(0.029)	(0.028)	(0.041)	(0.045)	(0.043)
Treated x ICT Share Investing IAB-LIAB (demeaned)	0.793**	0.710*	0.880**	-0.358	0.439
	(0.374)	(0.389)	(0.439)	(0.582)	(0.46)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.013	0.014	-0.032	-0.022	0.002
	(0.030)	(0.029)	(0.041)	(0.045)	(0.045)
Treated x ICT Investment Share ifo (demeaned)	0.745*	0.800*	1.081**	0.390	-0.322
	(0.435)	(0.418)	(0.426)	(0.682)	(0.720)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.015	0.011	-0.036	-0.012	-0.010
	(0.030)	(0.029)	(0.042)	(0.045)	(0.043)
Treated x ICT Capital Share EU KLEMS (demeaned)	1.984	1.775	2.877*	-1.084	1.591
	(1.371)	(1.420)	(1.677)	(2.201)	(2.082)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.001	0.024	-0.019	-0.030	0.015
	(0.033)	(0.031)	(0.044)	(0.044)	(0.048)
Treated x ICT Share CapEx U.S. Economic Census (demeaned)	1.378	1.113	1.514	-2.525	3.568*
	(1.388)	(1.276)	(1.378)	(1.988)	(2.142)
Observations	8,521	8,466	8,623	8,968	8,008

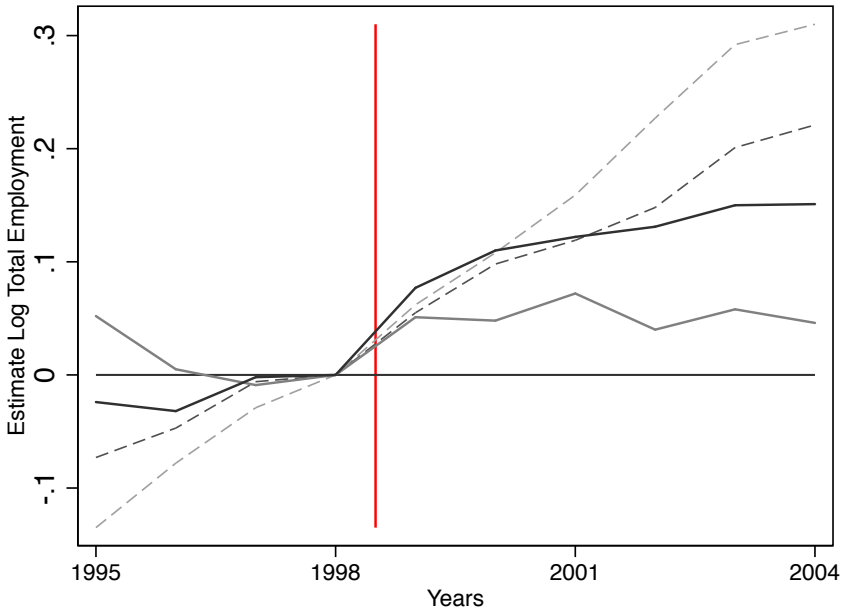
Note. Each coefficient is estimated from different regression following main specification (14) including interaction term. The dependent variables are the log of college educated vs. non-college educated in column (1), the log of full-time college educated vs. full-time non-college educated in column (2), the log of high-skilled occupations vs. low-skilled occupations in column (3), the log of manual occupations vs. service occupations in column (4) and the log of qualified manual occupations vs. unqualified manual occupations in column (5). The interaction term is the share of firms investing ICT in a given year based on the IAB-LIAB dataset in panel 1, the share of ICT investment based on Sauer, Strobel (2015) in panel 2, the ICT capital share for Germany based on EU KLEMS data and the ICT capital expenditure share based on U.S. Economic Census. Interaction terms are demeaned. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 12: Spillover Effects on Labor Inputs

	Dependent Variable: Log Total Employment		
	(1)	(2)	(3)
Firm size interval [40,225],[275,1500]			
Treated Firm	0.081**	0.088**	0.082**
	(0.036)	(0.036)	(0.036)
Regional Share Firms	0.272***	0.289***	0.253***
Below Cutoff	(0.095)	(0.090)	(0.084)
Observations	10,116	10,116	10,116
Firm size interval [30,225],[275,1500]			
Treated Firm	0.091**	0.097***	0.092***
	(0.035)	(0.035)	(0.035)
Regional Share Firms	0.188**	0.200**	0.171**
Below Cutoff	(0.091)	(0.090)	(0.077)
Observations	13,156	13,156	13,156
Firm size interval [20,225],[275,1500]			
Treated Firm	0.109***	0.114***	0.110***
	(0.037)	(0.038)	(0.037)
Regional Share Firms	0.148*	0.162*	0.138*
Below Cutoff	(0.087)	(0.084)	(0.072)
Observations	19,184	19,184	19,184
Firm size interval [10,225],[275,1500]			
Treated Firm	0.118***	0.119***	0.121***
	(0.039)	(0.038)	(0.037)
Regional Share Firms	0.054	0.058	0.041
Below Cutoff	(0.078)	(0.075)	(0.066)
Observations	34,019	34,019	34,019
Fixed Effects	Firm, Year	Firm, Industry-Year	Firm, Industry-Year, Area-Year

Note. Each coefficient is estimated from different regression following specification 16. The dependent variable is log total employment. Additional controls are log average firm wage on fixed effects specified in each column. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Figure 11: Dynamic Effect of Spillovers on Labor Inputs



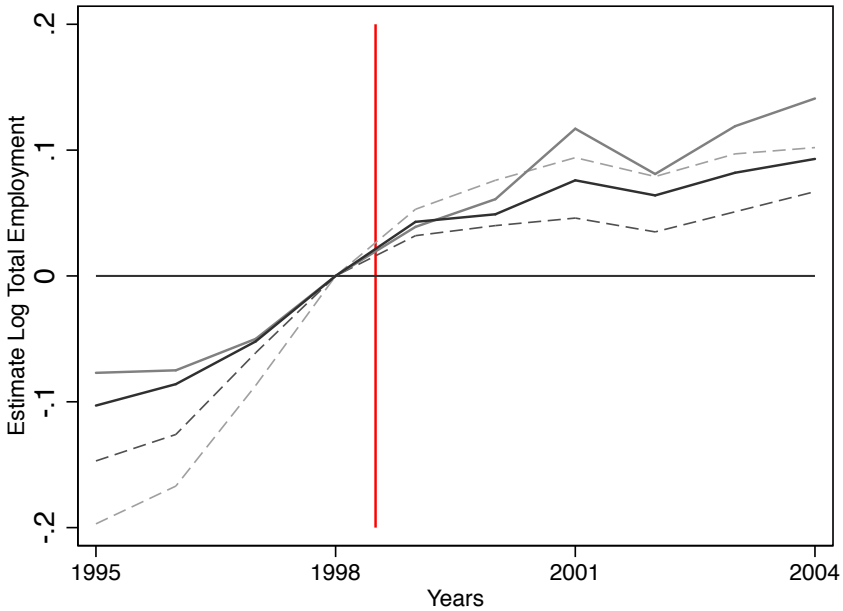
Note. Coefficients from estimation of dynamic specification (17). The dependent variable is the log of total employment. The light grey dashed line is for firm size interval [40,225],[275,1500], the dark dashed line for [30,225],[275,1500], the dark solid line for [20,225],[275,1500] and the light solid line for [10,225],[275,1500]. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and area (Bundesland)-year fixed effects.

Table 13: Spillover Effects on Labor Inputs

	Dependent Variable: Log Total Employment			
	(1)	(2)	(3)	(4)
Treated Firm	0.080**	0.088**	0.102***	0.105***
	(0.036)	(0.035)	(0.037)	(0.038)
Regional Share Weighted	0.196***	0.129**	0.128**	0.143***
Receiving Input-Output	(0.068)	(0.063)	(0.052)	(0.041)
Observations	10,406	13,446	19,474	34,309

Note. Each coefficient is estimated from different regression following specification 16. The dependent variable is log total employment. The firm size interval is [40,225],[275,1500] in column (1), [30,225],[275,1500] in column (2), [20,225],[275,1500] in column (3) and [10,225],[275,1500] in column (4). Additional controls are log average firm wage, firm fixed effects and year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Figure 12: Dynamic Effect of Spillovers on Labor Inputs



Note. Coefficients from estimation of dynamic specification (17). The dependent variable is the log of total employment. The light grey dashed line is for firm size interval [40,225],[275,1500], the dark dashed line for [30,225],[275,1500], the dark solid line for [20,225],[275,1500] and the light solid line for [10,225],[275,1500]. Additional controls are log average firm wage, firm fixed effects and year fixed effects.

Appendices

A Competing Programs

I interpret the results of the analysis as the effect of the investment tax credit program described in Section 1. As noted before, during the period of analysis further support was given through similar programs directed to firms in East Germany. These programs might confound the results, complicating statements with respect to tax credits and effect size. I describe the two most important competing programs and their differences with the tax credits. I explain how my estimation strategy that relies simultaneously on cross-sectional and time series variation controls for various confounding factors. I also provide additional robustness tests that suggest little influence if any from the competing programs. Taken together, the evidence supports the claim that the results in the main text are caused by the tax credits.

A.1 Special Depreciation Allowances

The German government introduced special depreciation allowances to all firms in East Germany in 1991.[40] The policy allowed to depreciate 50% of an asset's value directly after purchase but latest 5 years thereafter. The remaining value followed standard depreciation practices which was commonly done by equal amounts each year over the life span of an asset. Independent of the depreciation schedule, the accumulated depreciation of an asset are equal to the initial value. However, the additional allowance at the beginning shifts the tax benefits from depreciation towards earlier years. This reduces investment costs since discounting makes a current benefit more valuable than an equally sized future one.

Given that there is only a shift in timing, the seemingly large special depreciation allowance converts to a smaller capital cost reduction. House and Shapiro (2008) calculate the capital cost reduction for a 50% special depreciation allowance under U.S. regulations. They find that for short-lived assets (5 years), the program leads to a reduction in capital costs by 1.26%. For long-lived assets (20 years), the capital cost reduction is roughly

[40] The program is very similar to those studied in House and Shapiro (2008), Maffini et al. (2016) and Zwick and Mahon (2017). Eichfelder and Schneider (2014) analyze the German program comparing firm in East Germany with those in West Germany.

5%. Since many machinery goods have an estimated life time of maximum 10 years, the allowances provided relatively small benefits compared to the investment tax credit program.

The special depreciation program lasted until the end of 1998 coinciding with the change in the tax credit policy. However, all firms received the same special depreciation allowances and the program was discontinued for all firms at the same time. Thus, the program did not induce any differential treatment between small and large firms. I include time fixed effects in my estimation, which cover this program change.[41]

Nevertheless, firms from different industries likely differ in their capital structure, particularly in the life span of their employed assets. This leads to a different capital cost reductions from special depreciation allowances, with industries with longer-lived assets receiving larger reductions. If the capital structure of industries correlates with firm size, special depreciation allowances can influence treatment along the same dimensions as the tax credit program. However, since this type of influence is industry-specific, the use of yearly industry fixed effects can control for this type of influence.[42]

The simultaneity of the changes in both tax policies was not a coincidence. The special depreciation allowances were stopped with the goal of incorporating similar capital cost reductions through tax credits. This is the reason for the increase in the height of tax credits for all firms in 1999. At the same time, the policy shifted the focus towards small firms, introducing the additional differential treatment along firm size. There is no evidence that small firms had a different pre-policy trend or that policy makers anticipated differential shocks by firm size in the future. The focus on small firms seems to reflect a general preference for small firms which is a common occurrence for policies worldwide. Arguments brought forward are the large share of small firms in an economy and unfair competition advantages of large firms.

[41] I consider the differential treatment from adding the same tax credit rate to an already differentially distorted capital cost as negligible.

[42] Any remaining interactions between the two programs should not hamper the qualitative results assuming that special depreciation allowances were less important since both programs were pure investment cost shocks. However, since the exact size of the shock would be unclear, it would be difficult to state the effects in terms of an elasticity.

A.2 German Joint Task Program

The German Joint Task Program was another program through which firms in East Germany could receive a reduction in investment costs. The program already existed before reunification of Germany and is currently the main place-based policy instrument in Germany.[43] After reunification, the main focus was put on regions in East Germany. The program distributed funds to each state in Germany for infrastructure investments and firm-specific investments. Given general rules, each state then decided on more specific rules for the allocation of funds. Since funds were limited, decisions on allocation was based on a competitive basis. Important criteria were the expected success of firms and the commitment to keep employees or hire new workers. Thus, firms had uncertainty about the success of their application. The application also had to be done before initiating the investment. Thus, this program mixed a reduction in capital costs with a selection on firm characteristics and specific employment decisions. This in turn also led to various uncertainty about the success and the height of the grant.

Overall, during the years in question the joint task program was not as important as tax credits. Using information from the IAB-establishment survey for 1997, 76% single-establishment firms state that they used direct investment grants, whereas 34% used GRW grants. Most firms with GRW grants used tax credits at the same time as well. The amounts of funds distributed within East German manufacturing firms were roughly half of the tax credits. The program also focused on large investment projects with a priority for the opening of new firms.

Similar to the tax credits, the program provided additional incentives for small firms. However, the differential treatment stayed rather stable over time, with constant maximum subsidy rates for small and large firms respectively and little evidence for changes in applying this maximum subsidy rate. Thus, comparing small and large firms before and after the end of 1998 should identify the direct investment grant independent of the joint task program.

To further verify my claims, I consider a robustness test. In contrast to the tax credits, the GRW grants had preferential rates for small firms for structure investments.

[43] Dettmann et al. (2016) analyze the program using regional variation in West Germany and applying RDD techniques. Stierwald and Wiemers (2003) and Bade and Alm (2010) look at the program using firm-level variation.

Table A.1: Difference-in-Differences - Effect on Investment (AFID)

	Dependent Variable:		
	$\frac{\text{Total Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Equip Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Structure Invest}_t}{\text{Avg Tot Invest}}$
	(1)	(2)	(3)
Treatment	0.167**	0.174**	-0.007
	(0.080)	(0.067)	(0.031)
Observations	15,865	15,865	15,865

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are total investment over the average total investment between 1995-2004 in column (1), equipment investment over the average total investment between 1995-2004 in column (2) and structures investment over the average total investment between 1995-2004 in column (3). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table A.1 compares the effect of structures investments in comparison to equipment investments for manufacturing firms in East Germany. I use an investment measure that combines extensive and intensive margin effects. The first two columns are equivalent to the main text and show that the policy change increased scaled total investment by 0.167 and scaled equipment investment by 0.174. However, the effect of -0.007 for scaled structures investment is economically and statistically insignificant. This is in line with the regulations of the investment tax credit program but would be surprising for the GRW program.

B Theoretical Framework

The theoretical framework provides intuition for the empirical analysis. The theoretical framework section in the main text shows only the major steps. This appendix section provides more detail about calculation steps.

The model focuses on firm production in a partial equilibrium framework. I focus on heterogeneous firms i with each producing one differentiated good. These firms are spread throughout many regions r. The production function for each firm is

$$F(K_i, U_i, S_i) = Y_i = A_i A_r \left[(a_K K_i^{\rho} + a_S S_i^{\rho})^{\frac{\mu}{\rho}} + a_U U_i^{\mu} \right]^{\frac{1}{\mu}}, \tag{18}$$

where output Y_i is produced using capital K_i, low-skilled labor U_i and high-skilled labor S_i as input factors. The nested CES-production function is governed by the parameters a_K, a_S and a_U. The elasticity of substitution between low-skilled workers and capital (or high-skilled workers) is $\frac{1}{1-\mu}$ and the elasticity of substitution between high-skilled and

capital is $\frac{1}{1-\rho}$. Production also depends on a firm-specific production parameter A_i and a region-specific productivity shifter A_r.

The nesting of the three input factors follows Krusell et al. (2000). Importantly, the nesting allows for differential adjustment of each labor input. There is capital-skill complementarity when $\mu > \rho$. The additional regional-specific productivity shifter follows Greenstone et al. (2010) and Gathmann et al. (2017) and captures the effect of agglomeration economies.

Each firm maximizes their profit facing the downward sloping inverse demand curve

$$p_t = BY_t^{-\frac{1}{\eta^D}}, \tag{19}$$

and capital cost r, wage to low-skilled labor w_U and wage to high-skilled labor w_S. I assume perfect mobility of capital and labor leading to equalization of factor prices between regions. Each firm can receive a tax credit τ on their investments. Thus, the ultimate capital costs are firm specific since they include tax credits that differ from firm to firm.

Thus the firm maximization problem is

$$\max_{K_i,S_i,U_i} p_iY_i - (1-\tau_i)rK_i - w_SS_i - w_UU_i \tag{20}$$

The first order conditions from the maximization problem are

$$(1-\tau_i)r = \left(1-\frac{1}{\eta^D}\right) Ba_KY_i^{1-\mu-\frac{1}{\eta^D}} X^{\frac{\mu-\rho}{\rho}} K_i^{\rho-1}(A_iA_r)^{\mu} \tag{21}$$

$$w_S = \left(1-\frac{1}{\eta^D}\right) Ba_SY_i^{1-\mu-\frac{1}{\eta^D}} X^{\frac{\mu-\rho}{\rho}} S_i^{\rho-1}(A_iA_r)^{\mu} \tag{22}$$

$$w_U = \left(1-\frac{1}{\eta^D}\right) Ba_UY_i^{1-\mu-\frac{1}{\eta^D}} U_i^{\mu-1}(A_iA_r)^{\mu} \tag{23}$$

where $X_i = a_KK_i^{\rho} + a_SS_i^{\rho}$.

The FOCs serve to study how firms adjust their production process with tax credits. Considering tax credits as a form of reduction in capital costs, it is useful to show that

$$d\ln c_K = \frac{d(1-\tau_i)r}{(1-\tau_i)r} = -\frac{rd\tau_i}{(1-\tau_i)r} = \frac{d\tau_i}{1-\tau_i}, \tag{24}$$

where c_K are capital costs of a firm.

With that in mind, the total derivative of Equation (21) with respect to the tax credit rate is

$$-1 = \left(1 - \mu - \frac{1}{\eta^D}\right) \frac{dY_i}{d\tau_i} \frac{1-\tau_i}{Y_i} + \frac{\mu - \rho}{\rho} \frac{dX_i}{d\tau_i} \frac{1-\tau_i}{X_i} + (\rho - 1) \frac{dK_i}{d\tau_i} \frac{1-\tau_i}{K_i} + \mu \frac{dA_r}{d\tau_i} \frac{1-\tau_i}{A_r} \quad (25)$$

To simplify calculations of Equation (22), first I divide left and right side with Equation (21) and then take the total derivative to receive

$$\frac{dS_i}{d\tau_i} \frac{1-\tau_i}{S_i} = \frac{dK_i}{d\tau_i} \frac{1-\tau_i}{K_i} - \frac{1}{1-\rho} \quad (26)$$

The total derivative of Equation (23) is

$$\frac{dU_i}{d\tau_i} \frac{1-\tau_i}{U_i} = \left(1 - \frac{1}{(1-\mu)\eta^D}\right) \frac{dY_i}{d\tau_i} \frac{1-\tau_i}{Y_i} + \frac{\mu}{1-\mu} \frac{dA_r}{d\tau_i} \frac{1-\tau_i}{A_r} \quad (27)$$

The total derivatives depend on the elasticity of output and the elasticity of X with respect to tax credits.

$$\frac{dX_i}{d\tau_i} \frac{1-\tau_i}{X_i} = \rho a_K \frac{K^\rho}{X} \frac{dK_i}{d\tau_i} \frac{1-\tau_i}{K_i} + \rho a_S \frac{S^\rho}{X_i} \frac{dS_i}{d\tau_i} \frac{1-\tau_i}{S_i} \quad (28)$$

$$\frac{dY_i}{d\tau_i} \frac{1-\tau_i}{Y_i} = \frac{dA_r}{d\tau_i} \frac{1-\tau_i}{A_r} + \left(\frac{A_i A_r}{Y_i}\right)^\mu \left(a_U U_i^\mu \frac{dU_i}{d\tau_i} \frac{1-\tau_i}{U_i} + a_K X_i^{\frac{\mu-\rho}{\rho}} K_i^\rho \frac{dK_i}{d\tau_i} \frac{1-\tau_i}{K_i} + a_S X_i^{\frac{\mu-\rho}{\rho}} S_i^\rho \frac{dS_i}{d\tau_i} \frac{1-\tau_i}{S_i}\right) \quad (29)$$

To solve for each elasticity without interdependence, I plug in elasticities of X_i and S_i in Equation (25) to end up with

$$\frac{dK_i}{d\tau_i} \frac{1-\tau_i}{K_i} = \frac{1}{1-\mu} - \frac{\mu - \rho}{(1-\rho)(1-\mu)} \frac{a_S S_i^\rho}{a_K K_i^\rho + a_s S_i^\rho} + \left(1 - \frac{1}{(1-\mu)\eta^D}\right) \frac{dY_i}{d\tau_i} \frac{1-\tau_i}{Y_i} + \frac{\mu}{1-\mu} \frac{dA_r}{d\tau_i} \frac{1-\tau_i}{A_r} \quad (30)$$

With this result, the elasticity for U_i and Equation (29) we get the output elasticity of

$$\frac{dY_i}{d\tau_i} \frac{1-\tau_i}{Y_i} = \eta^D \frac{dA_r}{d\tau_i} \frac{1-\tau_i}{A_r} + \eta^D \left(\frac{A_i A_r}{Y_i}\right)^\mu X_i^{\frac{\mu}{\rho}} \left(1 - \frac{a_S S_i^\rho}{a_K K_i^\rho + a_S S_i^\rho}\right) \quad (31)$$

and then the elasticity of capital with respect to the rate of tax credits

$$\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} = \eta^D \left(\frac{A_iA_r}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + \frac{1}{1-\mu}\left(\frac{A_iA_r}{Y_i}\right)^{\mu}\left(a_U U_i^{\mu} + X_i^{\frac{\mu}{\rho}}\frac{a_S S_i^{\rho}}{X_i}\right) - \frac{\mu-\rho}{(1-\rho)(1-\mu)}\frac{a_s S_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (32)$$

We can rewrite this result as

$$\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} = \eta^D \left(\frac{A_iA_r}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + \frac{1}{1-\mu}\left(\frac{A_iA_r}{Y_i}\right)^{\mu} a_U U_i^{\mu}\frac{a_K K_i^{\rho}}{X_i} + \frac{1}{1-\rho}\frac{a_S S_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (33)$$

Then we get for high-skilled

$$\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i} = \left(\eta^D - \frac{1}{1-\mu}\right)\left(\frac{A_iA_r}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)\frac{a_K K_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (34)$$

We get for low-skilled

$$\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} = \left(\eta^D - \frac{1}{1-\mu}\right)\left(\frac{A_iA_r}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (35)$$

Also,

$$\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} = \left[\eta^D s_K + \frac{1}{1-\mu}s_U + \frac{1}{1-\rho}s_S + \left(\frac{1}{1-\rho} - \frac{1}{1-\mu}\right)s_U\frac{a_S S_i^{\rho}}{X_i}\right]H \quad + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (36)$$

$$\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i} = \left[\left(\eta^D - \frac{1}{1-\rho}\right)s_K + \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)s_U\frac{a_K K_i^{\rho}}{X_i}\right]H \quad + (\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \quad (37)$$

$$\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} = \underbrace{\left(\eta^D - \frac{1}{1-\mu}\right)s_K H}_{\text{direct effect}} \quad + \underbrace{(\eta^D - 1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r}}_{\text{indirect effect}}. \quad (38)$$

To consider changes in the composition of the labor force by skill type, I look at the

ratio between high-skilled and low-skilled. We have

$$\frac{S_i}{U_i} = \left(\frac{w_s}{w_u}\frac{a_u}{a_s}\right)^{\frac{1}{\rho-1}} X_i^{\frac{\rho-\mu}{\rho(\rho-1)}} U_i^{\frac{\rho-\mu}{1-\rho}} \tag{39}$$

The elasticity with respect to the tax credit can be calculated using the previous elasticities as

$$\frac{d\frac{S_i}{U_i}}{d\tau_i}\frac{1-\tau_i}{\frac{S_i}{U_i}} = \frac{\mu-\rho}{(1-\mu)(1-\rho)}\frac{a_K K_i^\rho}{X_i} \tag{40}$$

We can also consider a firm that does not receive a change in tax credit but which is in the same region as another firm that does. All steps from before can be redone taking into account that τ_j is not affected by changes in τ_i. This adjustment is factor neutral and we end up with

$$\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} = \frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j} = \frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} = (\eta^D-1)\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \tag{41}$$

and

$$\frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} = \eta^D\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} \tag{42}$$

Focusing now on the indirect effect through changes in the regional productivity shifter, I first relate the productivity shifter

$$A_r = \left(\sum Y_i\right)^\lambda. \tag{43}$$

to the total output in a specific region and the agglomeration elasticity λ.

Thus, the elasticity of regional productivity to a change in tax credits of one firm is

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \lambda\frac{d\sum Y_j}{d\tau_i}\frac{1-\tau_i}{\sum Y_j} = \lambda\frac{1-\tau_i}{\sum Y_j}\sum_j\frac{dY_j}{d\tau_i} \tag{44}$$

We can use the output elasticities calculated before. I consider that all but one firm adjust output because of the indirect channel only. For simplification I also assume that $\tau_i = \tau$ for all firms. Thus the elasticity of regional productivity is

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \frac{\lambda\eta^D}{1-\lambda\eta^D}\left(\frac{A_iA_r}{Y_i}\right)^\mu X_i^{\frac{\mu}{\rho}}\frac{a_K K_i^\rho}{X_i}\frac{Y_i}{\sum Y_j} = \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{(1-\tau)rK_i}{p_iY_i}\frac{\eta^D}{\eta^D-1}\frac{Y_i}{\sum Y_j}. \tag{45}$$

Since $\lambda > 0$ and $\eta^D > 1$, a scenario with $\lambda\eta^D > 1$ and therefore a negative elasticity is

possible. However, this scenario is a boundary case and does not have economic meaning.

If several firms receive a change in tax credits, then a sequential adjustment of one firm after the other is the most intuitive way of calculation. However, even though the assumption $\tau_i = \tau$ is not correct, I still maintain it. I also assume equal capital shares in all firms. With that we can add up all elasticities where there is a change in τ_i

$$= \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{(1-\tau)rK_i}{p_iY_i}\frac{\eta^D}{\eta^D-1}\frac{\sum Y_{i|d\tau_i\neq 0}}{\sum Y_j} \tag{46}$$

C Data

This section complements the Data section in the main text. I focus here on details of creating the final datasets, imputations of variables and specific sample selection. The following information pertains to the datasets by the Federal Statistical Office of Germany. The AFID Establishment-Panel[44], the Panel of Cost Structure[45] and the Establishment Panel.

C.1 Predicting Marginally Employed Workers in the BHP

For determining the firm size in context of the policy all employees count as one person independent of their employment contract or their working hours. This includes marginally employed workers as well.[46] The BHP records information about the number of marginally employed only since 1999 though. For classifying firms into small firms with fewer 250 or fewer employees and large firms with more than 250 employees, it is important to impute marginally employed for prior years.

I use the following approach. First, I calculate the share of marginally employed for each establishment averaging over the years 1999 and 2000. For establishments that existed in these years, I assume that this share is constant over time and impute the number of marginally employed for years before 1999 directly. For establishments that did not exist in 1999 or afterwards, this direct method does not work. For these cases, I

[44]Source: RDC of the Federal Statistical Office and Statistical Offices of the Länder, AFiD Establishment-Panel, 1995-2004, own calculations.

[45]Source: RDC of the Federal Statistical Office and Statistical Offices of the Länder, Panel of Cost Structure, 1999-2001, own calculations.

[46]The only exception are apprentices that have special contracts and therefore are not employees in legal terms.

run a simple prediction model. I include all establishments in 1999 and 2000 with more than 20 total employees. I run the model separately for East and West Germany and include 3-digit industry classifiers and firm size. I allow the coefficient for firm size to vary within each industry. Since the prediction model is linear, the predicted share can be negative. In these cases, I set the share to zero. I also winsorize at the 99th-percentile to exclude extreme values in the prediction. I then proceed as before and assume a constant share within each establishment.

Table C.1 provides information for the actual share, the calculated share and the predicted share.

C.2 Entry and Exit of Firms in the BHP

Since the difference-in-difference design makes use of the panel component, it is important to understand how firms enter and exit into the dataset. The BHP aggregates social security information on an establishment level and draws a 50% random sample of establishment identifiers. As noted in the data report by Schmucker et al. (2016), establishment identifiers can change when ownership changes or the establishment is merged with or split into different companies. Thus, administrative changes keeping the economic activity otherwise unaffected can be misclassified as entry or exit. To overcome this shortcoming, I merge an extension file provided by the IAB based on the methodology of Hethey and Schmieder (2010, 2013). In this file, entries and exits are classified according to flow information of coworkers from one establishment to another.[47] Essentially, if most employees move together from one establishment during its last year and constitute a large share of a newly created establishment, it likely constitutes a change only in the identifier. In contrast, if most workers move into unemployment or to various other establishments, it is likely a firm exited. Table C.2 provides information from the extension file for the overall dataset and the specific sample in the data analysis. ID changes do not happen that often. In the regression sample 0.59% of entering establishments and 0.6% exiting establishments are affected.

The dataset provides information on the preceding / succeeding establishment id for entering / exiting establishments with id change. I use this information to match these establishments. I take into account that the establishment id can change more than

[47] This methodology uses matched employer-employee data that is not accessible to me.

Table C.1: Marginally employed in BHP Data

Regression Sample	Mean	P10	P50	P90	Count
Actual Shares 1999 - 2004	0.045	0.000	0.013	0.098	9,271
Actual Shares 1999 - 2000	0.040	0.000	0.012	0.086	3,380
Imputed within establishment 1995-1998	0.040	0.000	0.012	0.083	6,516
Imputed with prediction model 1995-1998	0.052	0.000	0.051	0.106	231
Absolute Difference Actual Value vs Imputed within Establishment 2001-2004	0.027	0.000	0.009	0.055	5,890
Absolute Difference Actual Value vs Prediction Model 2001-2004	0.057	0.004	0.037	0.102	5,891

Note: Statistics are based on establishments in the manufacturing sector for the years 1995-2004 excluding Berlin. The number of observations is based on the statistic for number of employees.

Table C.2: Descriptive Statistics - IAB-BHP Data

ID Changes Entering Establishments	
Overall (%)	0.17
Regression sample (%)	0.35
Regression sample (count)	56
ID Changes Exiting Establishments	
Overall (%)	0.16
Regression sample (%)	0.48
Regression sample (count)	77
Matching Regression Sample	
Affected establishments (%)	8.23
Affected establishments (count)	148
Matched establishments (%)	5.92
Matched establishments (count)	109

Note:

once over the observation period. I case the information from id changes of entering and exiting establishments differs I choose the largest component. With this algorithm, I end up with 60 newly merged establishments in the regression sample which is a share of 3.24% of all establishments. There is also information on merges and split ups. However, these changes are more likely to influence production and I therefore consider them as distinct.

There is a second dimension to entry and exit that to the best of my knowledge has not been addressed in the official data description. It is concerned with the question whether an entry and exit into the dataset coincides with actual start and end of economic production. On various occasions, an establishment starts to exist with one or few employees and only in the next or a later year there is a jump to a distinctively larger number of employees. The same pattern in reverse happens before the exit of various establishments. Table C.3 summarizes the incident of these patterns. Looking at the percentiles of yearly employment growth, the median firm has zero growth. At the 10th and 90th percentile, growth rates are already large with a decrease by 72% and an increase by 154%. The smallest and largest growth rates are extreme with either a fall to zero employees or an increase of 36,800%. This is not the case for the AFID data. Interestingly, many of these changes coincide with movement of employees from or to other (not necessarily the same) employers. The share of inflowing employees that worked before is 72% during growth of

Table C.3: Descriptive Statistics

	Mean	Min	P10	P50	P 90	Max	Obs
Yearly employment growth rate	0.24	-1	-0.72	0.01	1.54	368	15,400
Yearly employment growth rate AFID	0.02	/	-0.11	0.01	0.15	/	15,867
Employment before increase by 200%		1	1	9	96	96	89
Employment after decrease by 67%		0	0	3	341	1,337	180
Share inflow from other establishments normal growth	0.39	0	0.0	0.38	0.71	1	14,802
Share outflow to other establishments normal growth	0.34	0	0.0	0.33	0.67	1	14,849
Share inflow from other establishments high growth	0.72	0	0.36	0.80	0.96	1	124
Share outflow to other establishments low growth	0.67	0	0.22	0.77	0.95	1	170

more than 200% but only 39% during lower growth excluding marginally employed. The share of outflowing employees that work afterwards is 67% during growth of less than -67% but only 34% during higher growth. This could potentially point to additional ID changes.

Because of these patterns, I assume that periods before growth of 200% or higher and periods after growth of -67% or lower are periods of non-production and I exclude them as observations.[48] These cutoffs are guided by establishment behavior in the AFID dataset. To provide an understanding how different cutoffs affect the empirical analysis, Table **??** provides estimates for employment using various cutoffs. The choice of the exact cutoff does not influence Figure **??** shows the dynamic profile.

[48] Although I consider it unlikely, it is possible, that the changes in employment happen because of large genuine adjustments in economic production from one year to another.

Table C.4: Static Estimation - Difference of Growth Rate Cutoff of Outliers

	1.5	2.0	2.5	3.0	5.0	10.0	none
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent Variable: Log(Number Employees)						
β	0.102***	0.076**	0.082**	0.080**	0.096**	0.063	0.125
	(0.029)	(0.030)	(0.035)	(0.035)	(0.046)	(0.052)	(0.089)
Obs	7,900	9,550	9,920	10,070	10,370	10,450	11,192
	Dependent Variable: Log(Ratio High-Skilled to Low-Skilled Occupation)						
β	-0.012	0.001	-0.005	-0.011	-0.020	-0.017	-0.030
	(0.033)	(0.031)	(0.033)	(0.032)	(0.029)	(0.030)	(0.031)
Obs	6,776	8,099	8,374	8,478	8,678	8,731	9,285

Note. Each coefficient is estimated from different regression following Specification 14. The dependent variables are machinery investment over lagged revenue in column (1), machinery investment over lagged imputed capital in column(2), the natural logarithm of machinery investment in column (3) and a dummy for positive machinery investment in column (4). The upper panel shows estimates for regressions including firms with 40 to 1,500 employees in 1998, the lower one for firms with 80 to 750 employees. Further, the sample consist of single-establishment manufacturing firms in East Germany excluding Berlin with observations in 1995 and 2004. For all regressions, firms with imputed negative capital are excluded. In Column (1) and (2), values are truncated at the largest 5%. Standard errors in parentheses are clustered at the firm level: * p<0.10, ** p<0.05, *** p<0.01

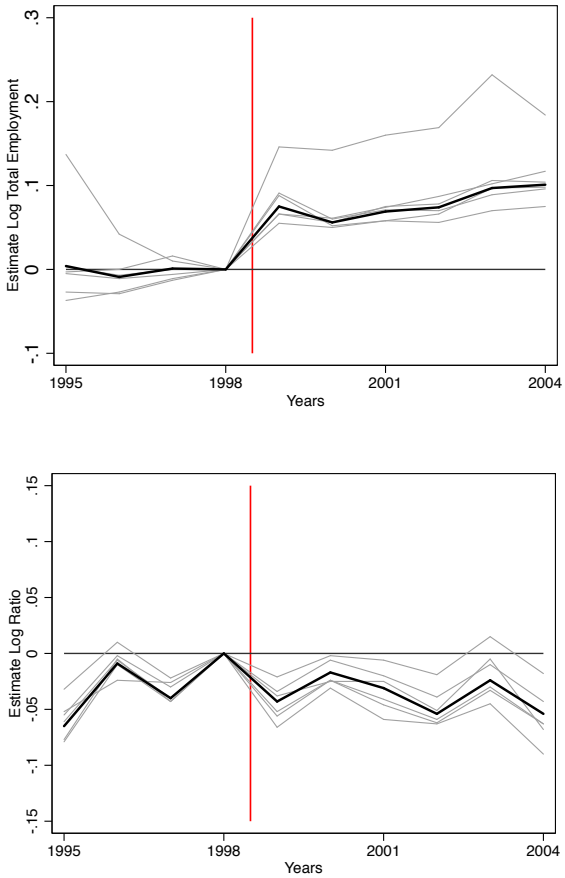


Figure 13: Dynamic Estimation - Difference of Growth Rate Cutoff of Outliers

Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total investment in the upper panel, log of equipment investment in the middle panel and a dummy for having positive total investment in the lower panel. Additional controls are log average establishment wage, establishment fixed effects, industry-year fixed effects and labor market-year fixed effects. Establishments with volatile investment, measured by growth of total investment in 1997 above the 95th percentile, are excluded. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.